MOG JEE Main Full Test - 2024 Revised

Total Time: 180 minutes | Total Marks: 300 | Total Questions: 75

GENERAL INSTRUCTIONS

The Question Paper consists of three parts (Physics, Chemistry and Mathematics) with a total of **75** questions.
Each subject is divided into 2 sections:
Section 1: Single Correct Type Questions (4 options, only one correct)
Section 2: Numerical Value Type Questions (answer to be entered as numerical value)
Marking Scheme:
Single Correct: +4 for correct, -1 for wrong, 0 if unattempted.
Numerical Value: +4 for correct, -1 for wrong.

By: C I P H Ξ R

Q.1

Assume temperature of $27^{\circ}\mathrm{C}, \mathrm{P} = 10^5\ \mathrm{Pa}, \mathrm{n} = 1$ moles, diameter of air molecules as $10^{-10}\ \mathrm{m}$, which of the following is correct?

(A) Mean free path > interatomic spacing

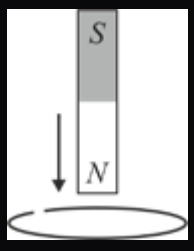
(B) Mean free path = interatomic spacing

(C) Mean free path < interatomic spacing

(D) Data insufficient

Q.2

A copper ring having a cut such as not to form a complete loop is held horizontally and a bar magnet is dropped through the ring with its length along the axis of the ring, figure. The acceleration of the falling magnet is



(A) g

(B) Less than g

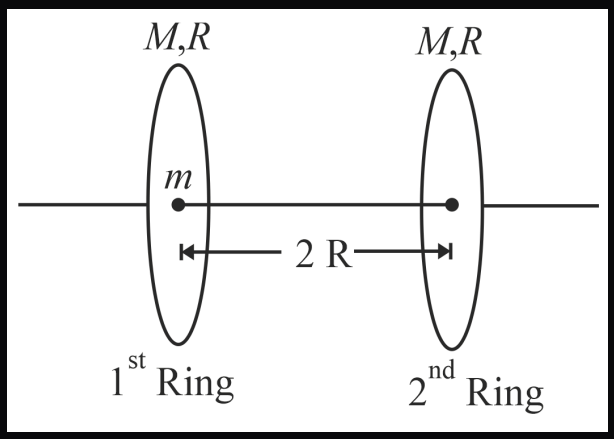
(C) More than g

(D) Zero

Q.3

Consider two identical rings of mass M and radius R fixed on a horizontal axis in a gravity-free space. A particle of

mass m is held at rest at the centre of 1st ring and projected along the axis towards 2nd ring. The minimum speed of projection so that the particle reaches the centre of 2nd ring is:



(A) $\sqrt{\frac{GM}{R}}$

(B) $\sqrt{\frac{2GM}{R}}$

(C) $\sqrt{\frac{2GM}{R}\left(1+\frac{1}{\sqrt{5}}-\sqrt{2}\right)}$

(D) $\sqrt{\frac{GM}{R}\left(1+\sqrt{5}-\frac{1}{\sqrt{2}}\right)}$

Q.4

Match the following (Take the relative strength of the strongest fundamental forces in nature as one)

I	II
Fundamental forces in nature	Relative strength
(A) Strong nuclear force	(e) 10^{-2}
(B) Weak nuclear force	(f) 1
(C) Electromagnetic force	(h) 10^{-13}
(D) Gravitational force	(i) 10^{-39}

The correct match is

(A) $\mathrm{A-f, B-i, C-e, D-h}$

(B) $\mathrm{A-f, B-h, C-e, D-h}$

(C) $\mathrm{A-f, B-h, C-e, D-i}$

(D) $\mathrm{A-f, B-e, C-h, D-i}$

Q.5

Which, among the following, is a correct statement?

(A) binding energy of a nucleus is always negative

(B) binding energy of a nucleus may be positive

(C) higher value of binding energy per nucleon means the nucleus is more ununstable

(D) higher value of binding energy per nucleon means the nucleus is more stable

Q.6

A plane electromagnetic waves travelling along the x-direction has a wavelength of $3\ \text{mm}$. The variation in the electric field occurs in the y-direction with an amplitude $66\ \text{Vm}^{-1}$. The equation for the electric and magnetic fields as a function of x and t are respectively

(A) $E_y = 33\cos\pi \times 10^{11}\left(t - \frac{x}{c}\right)$ and

$B_z = 1.1 \times 10^{-7}\cos\pi \times 10^{11}\left(t - \frac{x}{c}\right)$

(B) $E_y = 11\cos 2\pi \times 10^{11}\left(t - \frac{x}{c}\right)$ and

$B_y = 11 \times 10^{-7}\cos 2\pi \times 10^{11}\left(t - \frac{x}{c}\right)$

(C) $E_x = 33\cos\pi \times 10^{11}\left(t - \frac{x}{c}\right)$ and

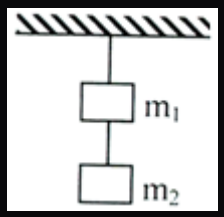
$B_y = 11 \times 10^{-7}\cos\pi \times 10^{11}\left(t - \frac{x}{c}\right)$

(D) $E_y = 66\cos 2\pi \times 10^{11}\left(t - \frac{x}{c}\right)$ and

$B_z = 2.2 \times 10^{-7}\cos 2\pi \times 10^{11}\left(t - \frac{x}{c}\right)$

Q.7

Two wires of same material are hanging as shown in the figure. Breaking stress is equal for both the wires.



Assertion: If mass 'm_2' increases, then upper wire will break first.
Reason: Breaking load of two wires of same material can be different.

(A) Assertion is true, reason is true; reason is a correct explanation for assertion.

(B) Assertion is true, reason is true; reason is not a correct explanation for assertion.

(C) Assertion is true, reason is false.

(D) Assertion is false but, reason is true.

Q.8

Assertion: Alkali metals are most suitable for photoelectric effect.
Reason: The work function of alkali metals is small, so even the radiations of small energy are able to cause the photoelectric effect.

(A) Assertion is True, Reason is True; Reason is a correct explanation for Assertion

(B) Assertion is True, Reason is True; Reason is not a correct explanation for Assertion

(C) Assertion is True, Reason is False

(D) Assertion is False but, Reason is True.

Q.9

Every atom makes one free electron in copper. If $1.1\ \mathrm{A}$ Current is flowing in the wire of copper having $1\ \mathrm{mm}$ diameter, then the drift velocity (approx.) will be (density of copper $= 9 \times 10^3\ \mathrm{kg\ m^{-3}}$ and atomic weight of copper $= 63$)

(A) $0.1\ \mathrm{mm\ s^{-1}}$

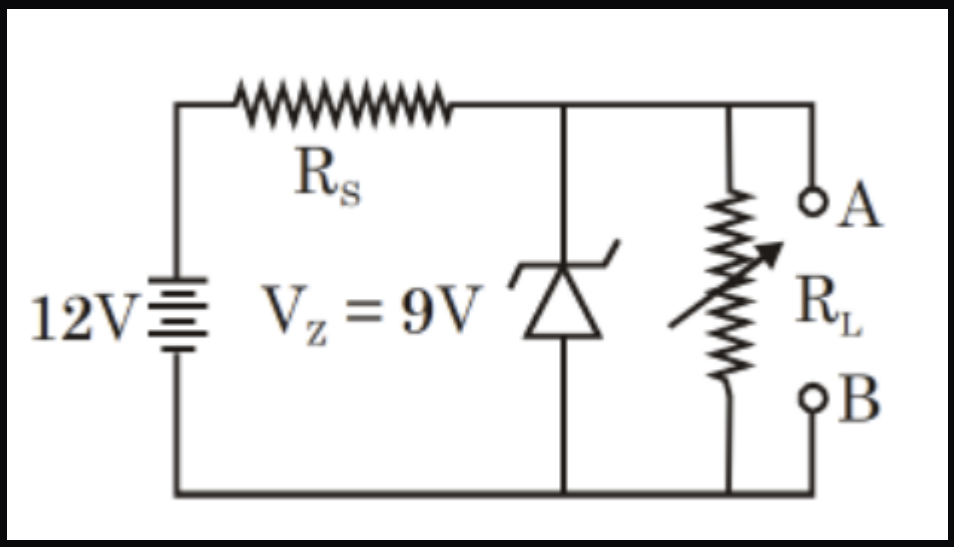
(B) $0.2\ \mathrm{mm\ s^{-1}}$

(C) $0.3\ \mathrm{mm\ s^{-1}}$

(D) $0.5\ \mathrm{mm\ s^{-1}}$

Q.10

Drawing power from a $12\ \mathrm{V}$ car battery, a $9\ \mathrm{V}$ stabilized DC voltage is required to power a car stereo system, attached to the terminals A and B, as shown in the figure. If a Zener diode with ratings, $\mathrm{V_z} = 9\ \mathrm{V}$ and $\mathrm{P_{max}} = 0.27\ \mathrm{W}$, is connected as shown in the figure, for the above purpose, the minimum series resistance R_s must be :-



(A) 111Ω

(B) 103Ω

(C) 100Ω

(D) 97Ω

Q.11

Two identical containers A and B having the same volume of an ideal gas at the same temperature have the mass of

the gas as m_1 and m_2 respectively and $2m_1 = 3m_2$. The gas in each cylinder expands isothermally to double of its volume. If the change in pressure in A is $300\ \mathrm{Pa}$, then the change in pressure in B is

(A) $200\ \mathrm{Pa}$

(B) $300\ \mathrm{Pa}$

(C) $400\ \mathrm{Pa}$

(D) $500\ \mathrm{Pa}$

Q.12

A diffraction pattern is obtained using a beam of red light. What happens if the red light is replaced by blue light?

(A) Diffraction bands become narrower and crowded together.

(B) Diffraction bands become broader and farther apart.

(C) Bands disappear

(D) Diffraction band firstly become narrower and then become broader.

Q.13

The actual length of a line is 5.32 m. First person take the measurement of the line as 4.23 and another person take the measurement is 4.9

(A) First person is more accurate less precise

(B) second person is less accurate more precise

(C) first person is less accurate more precise

(D) None of the above

Q.14

A car is moving along a straight road with uniform acceleration. It passes through two points P and Q separated by a distance with velocities $30\ \mathrm{kmh}^{-1}$ and $40\ \mathrm{kmh}^{-1}$ respectively. The velocity of car midway between P and Q is

(A) $33.3\ \mathrm{kmh}^{-1}$

(B) $1\ \mathrm{kmh}^{-1}$

(C) $25\sqrt{3}\ \mathrm{kmh}^{-1}$

(D) $35.35\ \mathrm{kmh}^{-1}$

Q.15

Two equal point charges are fixed at $x = -a$ and $\mathrm{x} = +\mathrm{a}$ on the x-axis. Another point charge Q is placed at the origin. The change in the electrical potential energy of Q, when it is displaced by a small distance x along the x-axis, is approximately proportional to $(x << \mathrm{a})$

(A) x

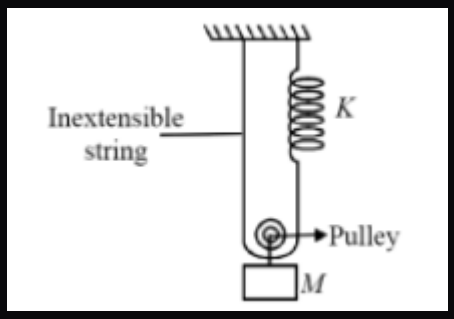
(B) x^2

(C) x^3

(D) $\frac{1}{x}$

Q.16

The time period of mass M when displaced from its equilibrium position and then released for the system as shown in figure is



(A) $2\pi\sqrt{\frac{M}{k}}$

(B) $2\pi\sqrt{\frac{M}{2k}}$

(C) $2\pi\sqrt{\frac{M}{4k}}$

(D) $2\pi\sqrt{\frac{2M}{k}}$

Q.17

The dimension $\left[\mathrm{M\,L^{-1}\,T^{-2}}\right]$ corresponds to,

(A) moment of force.

(B) surface tension.

(C) modulus of elasticity.

(D) coefficient of viscosity.

Q.18

A plate of mass m, length b and breadth a is initially lying on a horizontal floor with length parallel to the floor and breadth perpendicular to the floor. The work done to erect it on its breadth is

(A) $mg\left[\frac{b}{2}\right]$

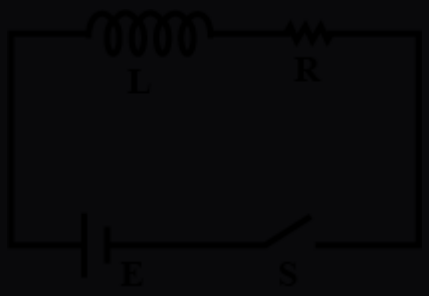
(B) $mg\left[a+\frac{b}{2}\right]$

(C) $mg\left[\frac{b-a}{2}\right]$

(D) $mg\left[\frac{b+a}{2}\right]$

Q.19

In the circuit shown in figure switch S is closed at time $t = 0$. The charge which passes through the battery in one time constant is



(A) $\frac{EL}{eR^2}$

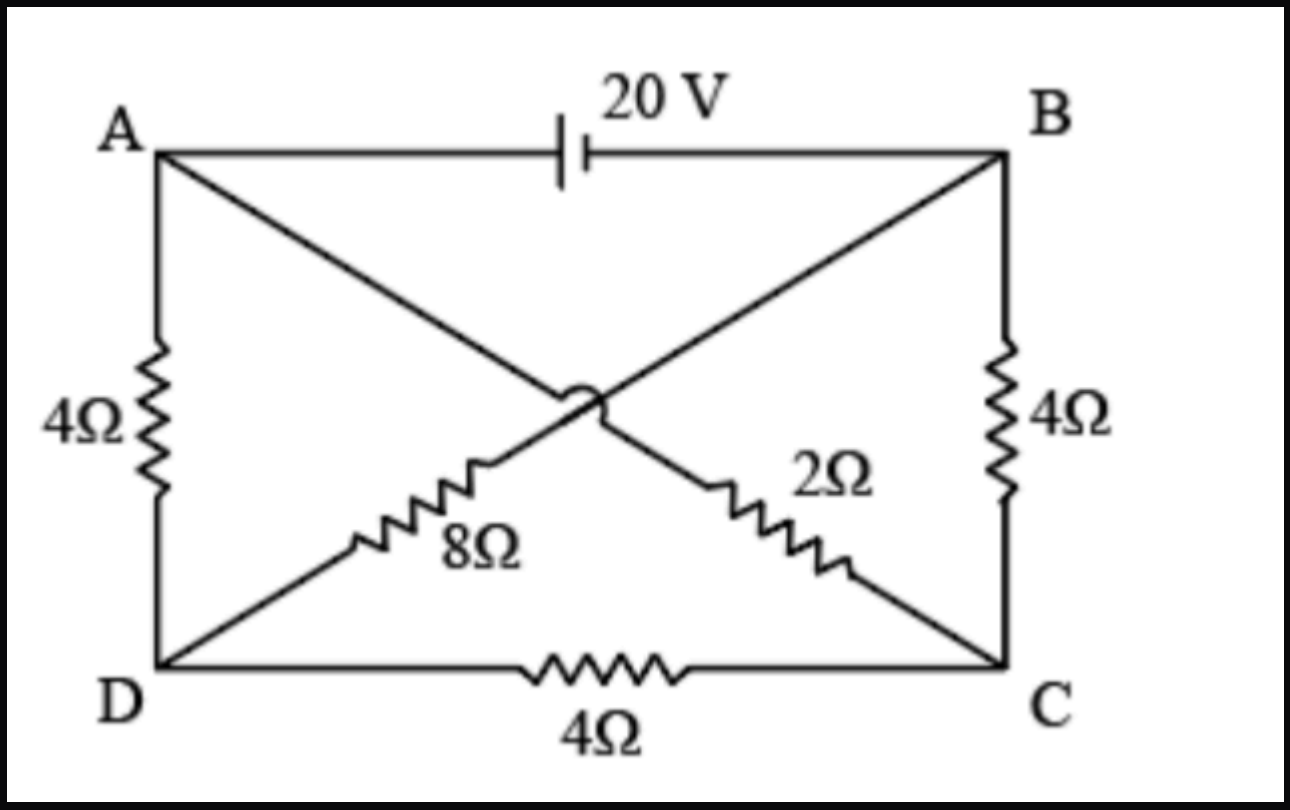
(B) $\frac{eL}{ER}$

(C) $\frac{eR^2E}{L}$

(D) $E\left(\frac{L}{R}\right)$

Q.20

Consider the circuit shown in the figure.



Which of the following statements are correct?

(A) Current through the battery is 5 A

(B) Current through branch AD is 2 A

(C) Current through branch BC is $3\ \mathrm{A}$

(D) The potential difference between points D and C is $4\ \mathrm{V}$

Q.21

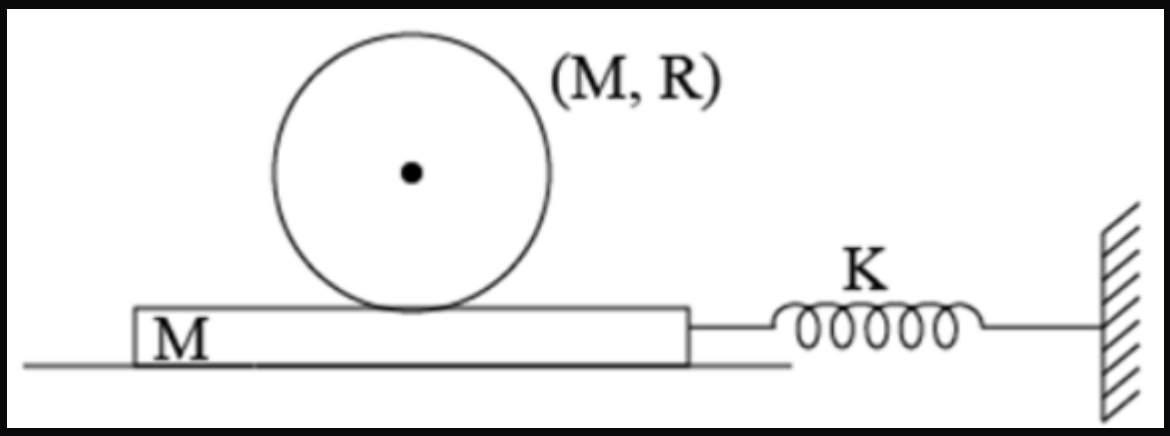
The speed of a wave on a string is $150\ \mathrm{m/s}$ when the tension is $120\ \mathrm{N}$. The percentage increase in the tension in order to raise the wave speed by 20% is (round off answer to nearest integer)

Q.22

A block of mass $m = 0.3\ \mathrm{kg}$ is connected to a spring of unknown spring constant k. It is compressed to a distance x from its equilibrium position and released from rest. After approaching half the distance $\left(\frac{x}{2}\right)$ from equilibrium position, it hits another identical block and combines to move with a common velocity $2ms^{-1}$. The total initial energy of the spring is E. find 10E.

Q.23

A cylinder of mass $M = 2\ \mathrm{kg}$ and radius $R = 12\ \mathrm{cm}$ lies on a plank of the same mass as shown in the figure. The surface between plank and ground is smooth but there is friction between cylinder and plank. If the coefficient of friction between the cylinder and the plank is $\mu = 0.4$, then what maximum initial compression (in cm) can be given to the spring such that the cylinder moves without slipping with respect to the plank? [Given, $k = 200\ \mathrm{N\ m^{-1}}$]



Q.24

A hydrogen-like atom (described by the Bohr model) is observed to emit six wavelengths, originating from all possible transitions between a group of levels. These levels have energies between $-0.85\mathrm{eV}$ and $-0.544\mathrm{eV}$ (including both these values). Find the atomic number of the atom. (Take $\mathrm{hc} = 1240\mathrm{eVnm}$, ground state energy of hydrogen atom $= -13.6\mathrm{eV}$)

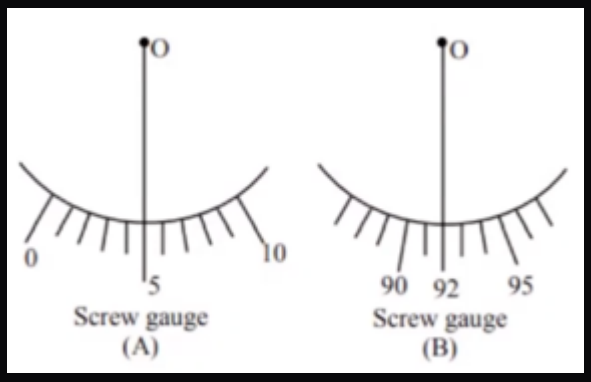
Q.25

Student A and student B used two screw gauges of equal pitch and 100 equal circular divisions to measure the radius of a given wire. The actual value of the radius of the wire is $0.322\ \mathrm{cm}$. The absolute value of the difference

between the final circular scale readings observed by the students A and B is ____.

[Figure shows position of reference O when jaws of screw gauge are closed]

Given pitch $= 0.1$ cm



Q.26

Trans$-2-$butene reacts with cold dilute $KMnO_4$ solution to produce:

(A) meso$-2,\ 3-$butanediol

(B) $(\pm) - 2-$butanol

(C) $(\pm)\ 2,\ 3-$butanediol

(D) $2,\ 3-$butanedione

Q.27

Alkyl cyanides undergo Stephen reduction to produce

(A) Aldehyde

(B) Secondary amine

(C) Primary amine

(D) Amide

Q.28

The strongest Lewis acid is -

(A) BF_3

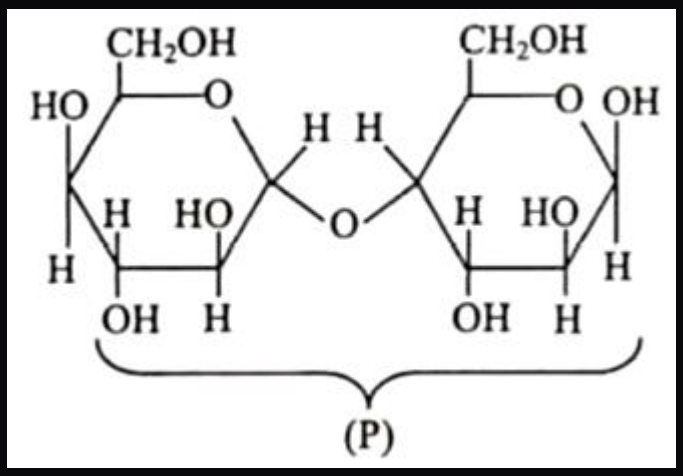
(B) BCl_3

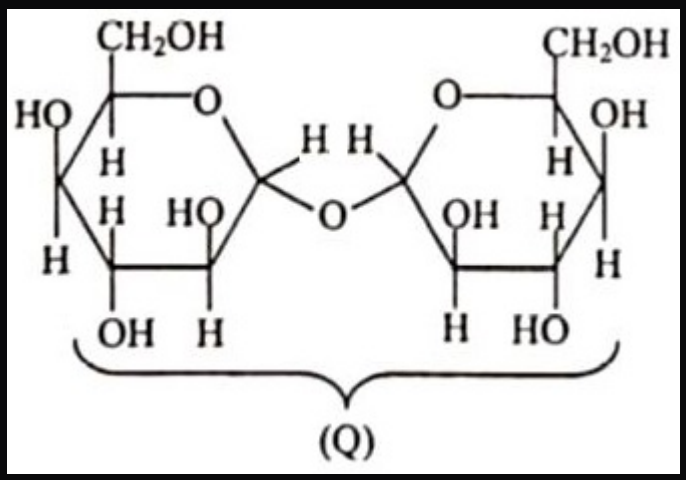
(C) BBr_3

(D) BI_3

Q.29

Which of the following disaccharide will not reduce Tollen's reagent?





(A) P

(B) Q

(C) P and Q both

(D) None of these

Q.30

The correct sequence of decreasing number of π-bonds in the structures of H_2SO_3, H_2SO_4 and $H_2\ S_2O_7$ is :

(A) $H_2SO_3 > H_2SO_4 > H_2\ S_2O_7$

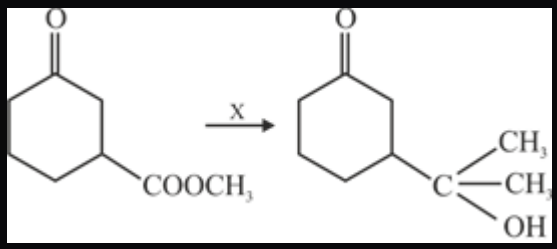
(B) $H_2SO_4 > H_2\ S_2O_7 > H_2SO_3$

(C) $H_2\ S_2O_7 > H_2SO_4 > H_2SO_3$

(D) $H_2\ S_2O_7 > H_2SO_3 > H_2SO_4$

Q.31

Consider the following reaction



The reagent ' X in the reaction is

(A) $\mathrm{HOCH_2-CH_2OH}\left|\mathrm{H^+, LiAlH_4}\right|$ ether, $\mathrm{2CH_3MgBr, H_3O^+}$

(B) $\mathrm{2CH_3MgBr}$ and $\mathrm{H_3O^+}$

(C) $\mathrm{HOCH_2-CH_2OH|H^+, 2CH_3MgBr, H_3O^+}$

(D) $\mathrm{HOCH_2-CH_2OH-|H^+, H_2|Pt, CH_3OH, H^+}$

Q.32

The red colour of oxyhaemoglobin is mainly due to

(A) d-d transition

(B) metal to ligand charge transfer transition

(C) ligand to metal charge transfer transition

(D) intraligand $\pi-\pi^*$ transition

Q.33

The correct order of ionic radii of $\mathrm{Ce, La, Pm}$ and Yb in $+3$ oxidation state is

(A) $Yb^{3+}<Pm^{3+}<Ce^{3+}<La^{3+}$

(B) $\mathrm{La^{3+}}<Ce^{3+}<\mathrm{Pm^{3+}}<Yb^{3+}$

(C) $Yb^{3+}<Ce^{3+}<Pm^{3+}<La^{3+}$

(D) None of these

Q.34

Assertion: Refining of petroleum involves the fractional distillation.

Reason: The fractional distillation involves repeated distillation. Repeated distillation means repeated heating.

(A) Both assertion and reason are true and the reason is the correct explanation of the assertion.

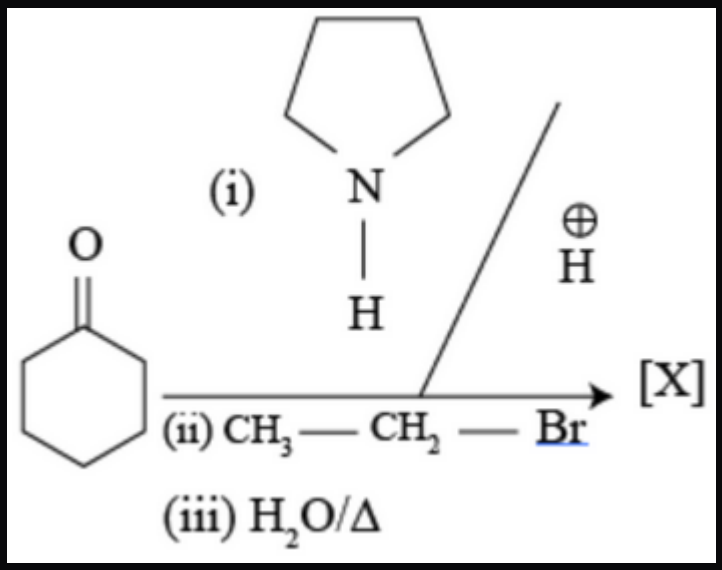
(B) Both assertion and reason are true, but the reason is not the correct explanation of the assertion.

(C) Assertion is true, but the reason is false.

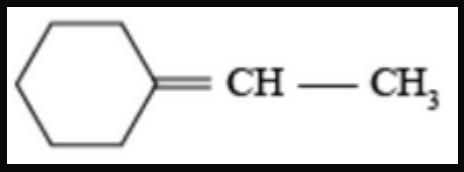
(D) The assertion and the reason both are false.

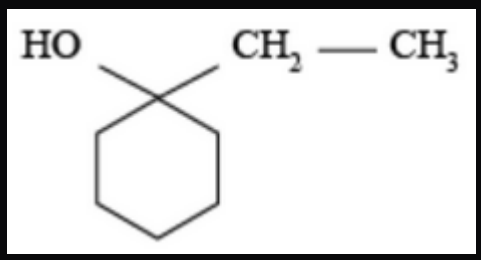
Q.35

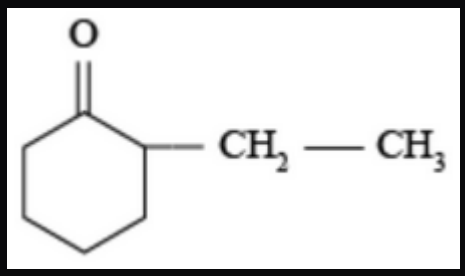
In the reaction sequence

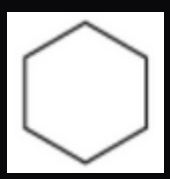


[X will be]

(A) 

(B) 

(C) 

(D) 

Q.36

The order of basic strength for methyl substituted amines in aqueous solution is

(A) $N(CH_3)_3 > N(CH_3)_2H > N(CH_3)H_2 > NH_3$

(B) $N(CH_3)H_2 > N(CH_3)_2H > N(CH_3)_3 > NH_3$

(C) $NH_3 > N(CH_3)H_2 > N(CH_3)_2H > N(CH_3)_3$

(D) $N(CH_3)_2H > N(CH_3)H_2 > N(CH_3)_3 > NH_3$

Q.37

Which of the following statement is true?

(A) Rest mass of photon is zero

(B) Plank constant has unit of energy

(C) Splitting of spectral lines in magnetic field is called stark effect

(D) Bohr's model is applicable to single electron species as well as multi electron species

Q.38

An aromatic amine (X) was treated with alcoholic potash and another compound (Y), then the foul smelling gas C_6H_5NC is formed. The compound (Y) was formed by reacting compound (Z) with Cl_2 in the presence of slaked lime. The compound (Z) is:

(A) $CHCl_3$

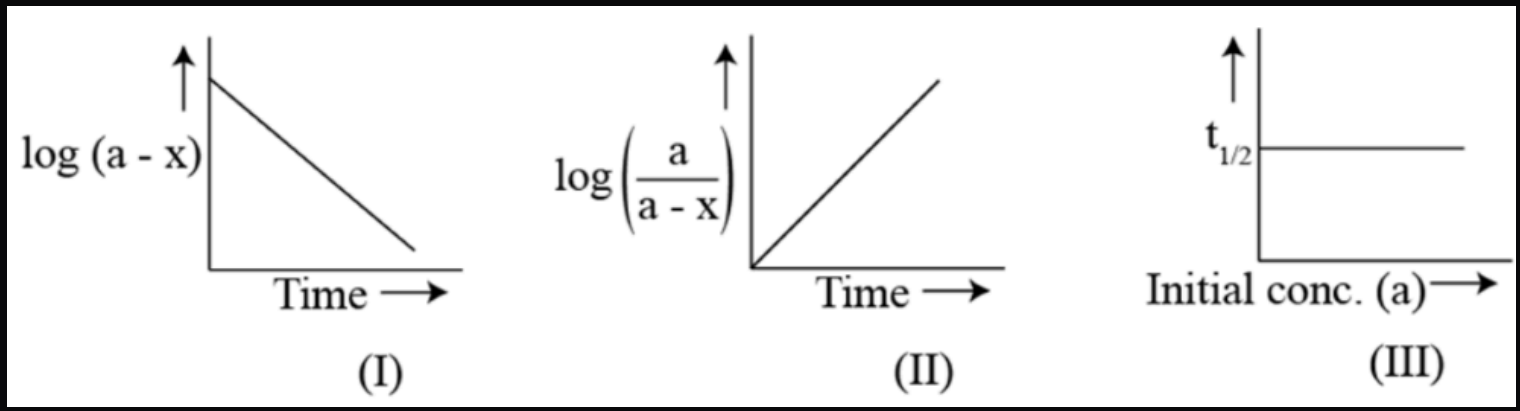
(B) CH_3COCH_3

(C) CH_3OH

(D) $C_6H_5NH_2$

Q.39

Which of the following is/are correct for the first order reaction? (a is initial concentration of reactant, x is concentration of the reactant reacted and t is time)



(A) I and II only

(B) II and III only

(C) I, II and III only

(D) I and III only

Q.40

The relationship between K_p and K_c is $K_p = K_c(RT)^{\Delta n}$. What would be the value of $\Delta \mathrm{n}$ for the reaction,

$\mathrm{NH_4Cl(s) \rightleftharpoons NH_3(\ g) + HCl(g)?}$

(A) 1

(B) 0.5

(C) 1.5

(D) 2

Q.41

Incorrect order of radius is

(A) $\mathrm{Sr^{2+} < Rb^{+} < Br^{-} < Se^{2-}}$

(B) $\mathrm{Nb^{5+} < Zr^{4+} < Y^{3+}}$

(C) $\mathrm{Co > Co^{2+} > Co^{3+} > Co^{4+}}$

(D) $\mathrm{Ba^{2+} > Cs^{+} > Se^{2-} > As^{3-}}$

Q.42

Experimentally it was found that a metal oxide has formula $\mathrm{M_{0.98}O}$. Metal M, is present as $\mathrm{M^{2+}}$ and $\mathrm{M^{3+}}$ in its oxide. Percentage of the metal which exists as $\mathrm{M^{3+}}$ would be:

(A) 4.08%

(B) 6.05%

(C) 5.08%

(D) 7.01%

Q.43

The pH of blood stream is maintained by a proper balance of $\mathrm{H_2CO_3}$ and $\mathrm{NaHCO_3}$ concentration. What volume of $5\mathrm{M}$ $\mathrm{NaHCO_3}$ solution should be mixed with $10\mathrm{ml}$ sample of blood which is $2\mathrm{M}$ in $\mathrm{H_2CO_3}$ in order to maintain a pH of 7.4 ($\mathrm{K_a}$ for $\mathrm{H_2CO_3}$ in blood is 4.0×10^{-7})?

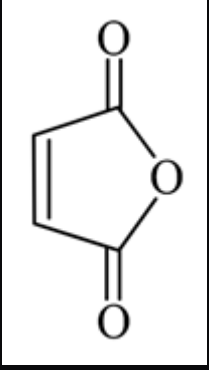
(A) $40\mathrm{ml}$

(B) $35\mathrm{ml}$

(C) $25\mathrm{ml}$

(D) $38\mathrm{ml}$

Q.44



Maleic anhydride

Maleic anhydride can be prepared by :

(A) Heating trans-but-2-enedioic acıd

(B) Heating cis-but-2-enedioic acid

(C) Treating cis-but-2-enedioic acid with alcohol and acid

(D) Treating trans-but-2-enedioic acid with alcohol and acid

Q.45

The CORRECT decreasing order of stability of the following carbocations is:

$\underset{\text{I}}{CH_3-\overset{\oplus}{C}H-CH_3}$ $\underset{\text{II}}{CH_3-\overset{\oplus}{C}H-O-CH_3}$ $\underset{\text{III}}{CH_3-\overset{\oplus}{C}H-COCH_3}$

(A) $\text{I} > \text{II} > \text{III}$

(B) $\text{III} > \text{II} > \text{I}$

(C) $\text{II} > \text{III} > \text{I}$

(D) $\text{II} > \text{I} > \text{III}$

Q.46

The number of geometrical isomers possible for the complex $[Pt\,(NH_3)\,(NH_2OH)\,(Py)Cl]^+$ is

Q.47

For a certain reaction : A(g) $\longrightarrow B(g)$

Half life for different initial pressures of A is given below:

P_{A_0} (atm)	0.1	0.025
$t_{1/2}$(sec)	100	50

If order of reaction is N. Find value of 4N.

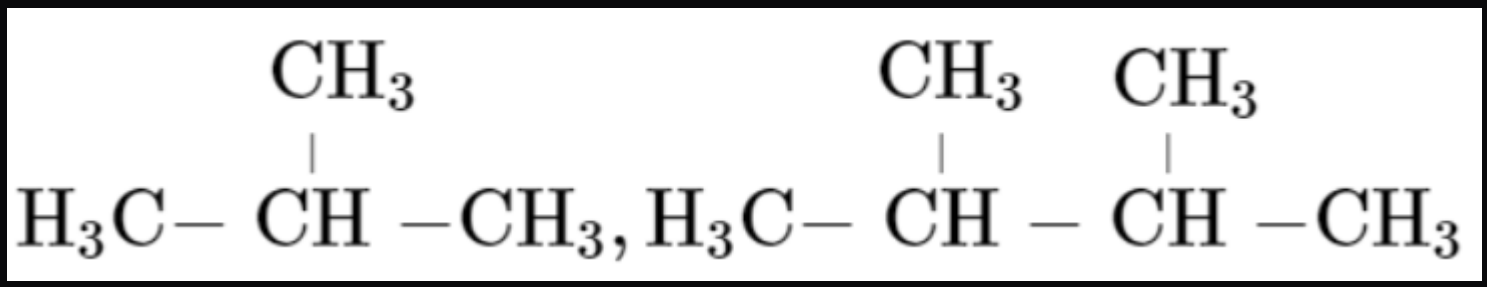
Q.48

If in $4f$ the number of radial nodes and angular nodes are X and Y respectively. Find the sum of $X + Y$?

Q.49

Consider the following compounds

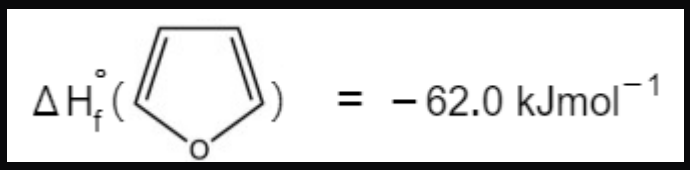
$CH_4, CH_3 - CH_3, CH_3OH, CH_3 - CH_2 - CH_3, CH_3 - CH_2 - CH_2 - CH_2 - CH_3$



how many alkanes can be prepared by wurtz reaction?

Q.50

Calculate heat of atomization of furan in kJmol^{-1} using the data



Heats of atomization of C, H, O are 717, 218, 249 kJmol^{-1} each isolated atom.

Q.51

Let $\vec{a} = 2\hat{i} + \hat{k}$, $\vec{b} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{c} = 4\hat{i} - 3\hat{j} + 7\hat{k}$. If $\vec{r}$ is a vector such that $\vec{r} \times \vec{b} = \vec{c} \times \vec{b}$ and $\vec{r} \cdot \vec{a} = 0$, then the value of $\vec{r} \cdot \vec{b}$ is

(A) 7

(B) -7

(C) -5

(D) 5

Q.52

If $A = \begin{bmatrix} 2 & 1 & -1 \\ 3 & 5 & 2 \\ 1 & 6 & 1 \end{bmatrix}$, then $\mathrm{tr}(\mathrm{Aadj}(\mathrm{adj}\, A))$ is equal to (where, $\mathrm{tr}(P)$ denotes the trace of the matrix P i.e. the sum of all the diagonal elements of the matrix P and $adj(P)$ denotes the adjoint of matrix P)

(A) 7

(B) 18

(C) -58

(D) -1624

Q.53

ABC is a variable triangle such that A is $(3,\ 2)$ and B and C lie on line $y = 2x + \lambda$ (where λ is a variable), then locus of the orthocenter of triangle ABC is-

(A) $x + 2y = 0$

(B) $x + 2y - 7 = 0$

(C) $(x-1)^2 + y^2 = 4$

(D) $x^2 + (y-1)^2 = 4$

Q.54

Let $a_n = 16, 4, 1, \ldots\ldots$ be a geometric sequence. The value of $\sum_{n=1}^{\infty} \sqrt[n]{P_n}$, where P_n is the product of the first n terms, is equal to.

(A) 8

(B) 16

(C) 32

(D) 64

Q.55

The least value of n for which $(n-2)x^2 + 8x + (n+4) > \sin^{-1}(\sin 12) + \cos^{-1}(\cos 12) \forall x \in R (n \in N)$ is

(A) 4

(B) 5

(C) 6

(D) 7

Q.56

A circle with centre at $(2, 4)$ is such that the line $x + y + 2 = 0$ cuts a chord of length 6 units. The radius of the circle is

(A) $\sqrt{41}$ units

(B) $\sqrt{11}$ units

(C) $\sqrt{21}$ units

(D) $\sqrt{31}$ units

Q.57

The number of 7 digit integers $abcdefg$, where $a < b < c < d > e > f > g$ such that $a, b, c, d, e, f, g \in \{1,2,3,\ldots,9\}$, are

(A) 700

(B) 20

(C) 720

(D) 800

Q.58

The locus of a point z which moves such that $2\arg\left(\frac{z-i+3}{z+3i-1}\right) = \pi$ is :

(A) a straight line passing through the points $(3-i)$ and $(-1+3i)$

(B) a straight line passing through the points $(-3+i)$ and $(1-3i)$

(C) a semicircle passing through the points $(-3-i)$ and $(1-3i)$

(D) a part of circle with centre at the point $(-1,-1)$ and radius $2\sqrt{2}$

Q.59

If $\{x\}$ denotes the fractional part of x then $\left\{\frac{3^{2n}}{8}\right\}$, $n \in N$ is

(A) $\frac{3}{8}$

(B) $\frac{7}{8}$

(C) $\frac{5}{8}$

(D) $\frac{1}{8}$

Q.60

The range of values of a, so that all the roots of the equation $2x^3 - 3x^2 - 12x + a = 0$ are real and distinct, belongs to:

(A) $(7, 20)$

(B) $(-7, 20)$

(C) $(-20, 7)$

(D) $(-7, 7)$

Q.61

There are n sets of observations given as $(1),\ (2\ ,3),\ (4\ ,5,6),\ (7\ ,8,9\ ,10),\ldots..$ The mean of the 13th set of observations is equal to

(A) 70

(B) 80

(C) 75

(D) 85

Q.62

Consider $f(x)=\sin x \forall x \in \left[0,\frac{\pi}{2}\right]; f(x)+f(\pi-x)=2 \forall x \in \left(\frac{\pi}{2},\pi\right]$ and $f(x)=f(2\pi-x) \forall x\ \in (\pi,2\pi)$. If n,m denotes number of points where $f(x)$ is discontinuous and non-differentiable respectively in $[0,2\pi)$, then value of $n+m$ is

(A) 0

(B) 1

(C) 2

(D) 4

Q.63

The perpendicular bisector of the line segment with endpoints $(2,3,2)$ and $(-4,1,4)$ passes through the point $(-3,6,1)$ and has equation of the form $\frac{x+3}{a}=\frac{y-6}{b}=\frac{z-1}{c}$ where $a,\ b$ and c are relatively prime integers with $a>0$. Find the value of $abc-(a+b+c)$:

(A) -1

(B) -2

(C) -3

(D) -4

Q.64

The value of $\sin\frac{9\pi}{14}\sin\frac{11\pi}{14}\sin\frac{13\pi}{14}$ is equal to

(A) $\frac{1}{64}$

(B) $-\frac{1}{64}$

(C) $\frac{1}{8}$

(D) $-\frac{1}{8}$

Q.65

Let $A = [a_{ij}]$ be a square matrix of order 3 and $B = [b_{ij}]$ be a matrix such that $b_{ij} = 2^{i-j}a_{ij}$ for $1 \leq i, j \leq 3,\ \forall i, j \in N$. If the determinant of A is same as its order, then the value of $\left|\left(B^T\right)^{-1}\right|$ is

(A) $\frac{1}{3}$

(B) 3

(C) 9

(D) $\frac{1}{27}$

Q.66

Shubham has 75% chance of attending the annual meet. Shikha has a 90% chance if Shubhan also attends otherwise she has a 40% chance of attending the meet. If I go to the annual meet and see Shikha there, then the probability that Shubham is also there, is

(A) $\frac{27}{31}$

(B) $\frac{19}{30}$

(C) $\frac{1}{5}$

(D) $\frac{9}{10}$

Q.67

Let $f\left(x\right) = \frac{x}{(1+x^n)^{\frac{1}{n}}}, n \geq 2,\ \ g\left(x\right) = \left(\underbrace{fofo\ldots.\text{ of}}_{n\ times}\right)\left(x\right)$ and $I(n,\ x) = \int g(x) \cdot x^{n-2}dx$, If $I\left(10, 1\right) = \frac{(p)^{\frac{m}{k}}}{q}$, then the value of $10\ k - m - q + p =$_____ (Given that G.C.D. $(m,\ k) = 1$))

(A) 11

(B) 10

(C) 8

(D) None of these

Q.68

If functions $f:\ (1, 2, \ldots,\ n) \to \{1995,\ 1996\}$ satisfying $f(1) + f(2) + \ldots + f(n) =$ odd integer are formed, then number of such functions can be:

(A) 2^n

(B) $2^{\frac{n}{2}}$

(C) n^2

(D) 2^{n-1}

Q.69

The solution of the differential equation $\frac{dy}{dx}+x(2x+y)=x^3(2x+y)^3-2$ is (C being an arbitrary constant)

(A) $\frac{1}{2x+xy}=x^2+1+Ce^x$

(B) $\frac{1}{(2x+y)^2}=x^2+1+Ce^{x^2}$

(C) $\frac{1}{2x+y}=x+1+Ce^{-x^2}$

(D) $\frac{1}{(2x+y)^2}=x^2+1+C$

Q.70

The sum of 50 observations and sum of their squares were found to be 245 and 1401 respectively, but later on, three observations 2, 3 and 5 were found to be not correct. If the incorrect observations are removed, then variance of remaining observations is

(A) 4.25

(B) 4.00

(C) 3.15

(D) 3.00

Q.71

If $\int \frac{x^{pq-p-1}}{(x^p+1)^q}dx=\frac{2(1+x^{-p})^{1-q}}{\lambda p(q-1)}+\text{c}\ \ (p,q\in N-\{1\})$, then the value of λ is (here, c is an arbitrary constant)

Q.72

The vertices of the triangle ABC are $A(0,0),\ B(3,0)$ and $C(3,4)$, where A and C are foci of an ellipse and B lies on the ellipse. If the length of the latus rectum of the ellipse is $\frac{12}{p}$ units, then the value of 2p is

Q.73

A continuous function $f(x)$ is such that $f(3x)=2f(x),\ \forall x\in R$. If $\int_0^1 f(x)dx=1$, then $\int_1^3 f(x)dx$ is equal to

Q.74

The area (in sq. units) bounded by $[x+y]=2$ with the co-ordinate axes is equal to A, find 2A (where, $[.]$ denotes the greatest integer function)

Q.75

$OABC$ is a tetrahedron in with O as the origin and position vectors of points $A,\ B,\ C$ as $\hat{i}+2\hat{j}+3\widehat{k},\ 2\hat{i}+\alpha\hat{j}+\widehat{k}$ and $\hat{i}+3\hat{j}+2\widehat{k}$ respectively, then the integral value of α to have shortest distance between $\overrightarrow{\text{OA}}$ & $\overrightarrow{\text{BC}}$ as $\sqrt{\frac{3}{2}}$, is

Answers & Solutions

Q.1 Answer:

—

Solution:

$$\lambda = \frac{\mathrm{v}}{\sqrt{2}\pi \mathrm{d}^2 \mathrm{nN}} = \frac{\mathrm{RT}}{\sqrt{2}\pi \mathrm{d}^2 \mathrm{pN}} \mathrm{a_a}$$

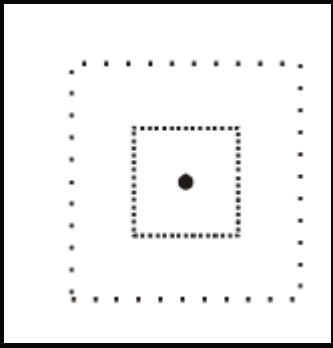
$$= \frac{\frac{25}{3} \times 300}{\sqrt{2} \times \pi \times 10^{-20} \times 10^5 \times 6 \times 10^{23}}$$

$$= \frac{25}{6\sqrt{2}\pi} \times 10^{-6}\ \mathrm{m}$$

$$\mathrm{x}^3 = \frac{\mathrm{v}}{\mathrm{nN_a}}$$

$$\Rightarrow \mathrm{x} = \left(\frac{\mathrm{RT}}{\mathrm{PN_a}}\right)^{1/3}$$

$$= \left(\frac{\frac{25}{3} \times 300}{10^5 \times 6 \times 10^{23}}\right)^{1/3}$$



Hence mean free path > interatomic spacing.

Q.2 Answer:

g

Solution:

Though emf is induced in the copper ring, but there is no induced current because of cut in the ring. Hence, nothing opposes the free fall of the magnet. Therefore, $a = g$.

Q.3 Answer:

$$\sqrt{\frac{2GM}{R}\left(1 + \frac{1}{\sqrt{5}} - \sqrt{2}\right)}$$

Solution:

The particle should be projected such that it can reach halfway because after midpoint it will move under the attraction of 2nd ring and will reach the centre of the ring.

$PE_i + KE_i = PE_{\text{middle}} + KE_{\text{middle}}$

$\Rightarrow \left(\frac{-GMm}{R} - \frac{GMm}{\sqrt{5}R}\right) + \frac{1}{2}mv^2 = \left(-\frac{GMm}{\sqrt{2}R} \times 2\right) + 0$

$\Rightarrow v = \sqrt{\frac{2GM}{R}\left[1 + \frac{1}{\sqrt{5}} - \sqrt{2}\right]}$

Q.4 Answer:

A – f, B – h, C – e, D – i

Solution:

I	II
Fundamental forces in nature	**Relative strength**
(A) Strong nuclear force	**(f)** 1
(B) Weak nuclear force	**(h)** 10^{-13}
(C) Electromagnetic force	**(e)** 10^{-2}
(D) Gravitational force	**(i)** 10^{-39}

So, A – f, B – h, C – e, D – i

Q.5 Answer:

higher value of binding energy per nucleon means the nucleus is more stable

Solution:

It has been observed that total mass of nucleus is always less than the sum of the masses of its nucleons. The energy difference between the nucleus and its constituent particles due to their mass difference is termed as the binding energy of the nucleus.

In other words, we can say that to break the nucleus into its constituent particles, some energy is needed to be supplied. This energy is termed as binding energy of the nucleus. More is the binding energy per nucleon, more is the energy required to break the nucleus and hence we can say the more stable the nucleus is.

Q.6 Answer:

$E_y = 66 \cos 2\pi \times 10^{11} \left(t - \frac{x}{c}\right)$ and

$B_z = 2.2 \times 10^{-7} \cos 2\pi \times 10^{11} \left(t - \frac{x}{c}\right)$

Solution:

Here, $\lambda = 3 \text{ mm} = 3 \times 10^{-3} \text{ m}, E_0 = 66 \text{ Vm}^{-1}$

$\therefore B_0 = \frac{E_0}{c} = \frac{66}{3 \times 10^8} = 2.2 \times 10^{-7} \text{ T}$

As, electromagnetic wave is propagating along x-axis and electric field oscillation is along y-direction, the magnetic field oscillation is along z-direction using the relation for harmonic wave

$E_y = E_0 \cos \frac{2\pi}{\lambda}(ct - x)$

$E_y = E_0 \cos \frac{2\pi c}{\lambda}(t - x/c)$

$\therefore E_y = 66 \cos \frac{2\pi \times 3 \times 10^8}{3 \times 10^{-3}}(t - x/c)$

$= 66 \cos 2\pi \times 10^{11}(t - x/d)$

and $B_z = B_0 \cos \frac{2\pi c}{\lambda}(\mathrm{t} - \mathrm{x/c})$

$= 2.2 \times 10^{-7} \cos 2\pi \times 10^{11}(t - x/c)$

Q.7 Answer:

Assertion is false but, reason is true.

Solution:

Given,

Breaking stress is equal for both the wires.

$\text{Breaking stress} = \frac{\text{Breaking load}}{\text{Cross-sectional area}}$.

Thus, the breaking load depends on the cross-sectional area of the wire if the breaking stress is same. Therefore, breaking load of two wires can be different and the lower wire may have much lesser breaking load as compared with upper wire.

Q.8 Answer:

Assertion is True, Reason is True; Reason is a correct explanation for Assertion

Solution:

Photoelectric effect is observable, only when an electromagnetic wave of certain maximum wavelength known as threshold wavelength or minimum frequency known as threshold frequency is incident on cathode plate.

It can be concluded from relation of energy of photon, $E = \phi = h\nu_o = \frac{hc}{\lambda_o}$**, where** $\phi,\ h,\ \nu_o,\ c \& \lambda_o$ **represents work function of metal plate, Plank's constant, threshold frequency, speed of light in vacuum and threshold wavelength.**

So, to have photoelectric effect energy of photon should provide at least energy equal to work function of metal, and also we know alkali metal has low work function, so these metals have maximum probability to emit photon even for the radiation having small amount of energy.

Q.9 Answer:

0.1 mm s^{-1}

Solution:

$v_d = \frac{i}{nAe'}$

where $n = \frac{N_A \rho}{M}$

$= \frac{6.023 \times 10^{26} \times 9 \times 10^3}{63} = 0.860 \times 10^{29}$

$= 8.6 \times 10^{28}$

and $A = \frac{\pi D^2}{4} = \frac{22}{7} \times \frac{\left(10^{-3}\right)^2}{4} \text{ m}^2$

$= \frac{11}{14} \times 10^{-6} \text{ m}^2$

$v_d = \frac{1.1}{8.6 \times 10^{28} \times \frac{11}{14} \times 10^{-6} \times 1.6 \times 10^{-19}}$

$= \frac{1}{9.6 \times 10^{+3}} = \frac{100 \times 10^{-4}}{96} = 1.0 \times 10^{-4} \text{ m s}^{-1}$

$= 0.1 \text{ mm s}^{-1}$

Q.10 Answer:

100Ω

Solution:

$\mathrm{i_{max}} = \frac{\mathrm{P_{max}}}{\mathrm{V_z}} = \frac{3}{100}$

$\therefore \mathrm{R_{S_{min}}} = \frac{\mathrm{V_{R_S}}}{\mathrm{i_{max}}}$

$= \frac{3}{3/100} = 100\Omega$

Q.11 Answer:

200 Pa

Solution:

$2m_1 = 3m_2$

$P_1 V = \frac{m_1}{M} RT$ and $P_2 V = \frac{m_2}{M} RT = \frac{2}{3} \frac{m_1}{M} RT$

$\frac{P_1}{P_2} = \frac{3}{2} \ldots\ldots \text{(i)}$

The process is isothermal $(P_1 V_1 = P_2 V_2)$

Hence, $\Delta P_1 = P_1 - \frac{P_1}{2} \Rightarrow P_1 = 2\Delta P_1$

$$\Delta P_2 = P_2 - \frac{P_2}{2} = \frac{P_2}{2} = \frac{1}{2}\frac{2}{3}P_1 = \frac{P_1}{3}$$
$$\Delta P_2 = \frac{2}{3}\Delta P_1 = \frac{2}{3}.300$$
$$\Delta P_2 = 200Pa$$

Q.12 Answer:

Diffraction bands become narrower and crowded together.

Solution:

The linear width of the central maxima in diffraction pattern is given by $\frac{2D\lambda}{a}$
where $\lambda =$ Wavelength
$a =$ Size of the slit
$D =$ Distance between slit and screen.
As $\lambda_{\text{Blue}} < \lambda_{\text{Red}}$ and width of diffraction bands is directly proportional to λ.
Hence, when the red light is replaced by blue light, the wavelength decreases which means that the fringe's width decrease and become narrower and crowded together.

Q.13 Answer:

first person is less accurate more precise

Solution:

The observation of first person is measured upto two points of decimal but it is far from the actual value. So, it is less accurate but more precise.

Q.14 Answer:

$35.35\ \text{kmh}^{-1}$

Solution:

$40^2 - 30^2 = 2\text{as}$ **, and** $v^2 - 30^2 = 2\text{a}\frac{s}{2}$
or $2\left(v^2 - 30^2\right) = 2as$
Comparing,
$2\left(v^2 - 900\right) = 1600 - 900 = 700$
or $v^2 = 900 + 350 = 1250$
or $v = 35.35\ \text{kmh}^{-1}$

Q.15 Answer:

x^2

Solution:

$\text{U}_\text{i} = \frac{2\text{kQq}}{\text{a}} + \frac{\text{kqq}}{2\text{a}}$

$\text{U}_\text{f} = \text{kQq}\left(\frac{1}{\text{a}+\text{x}} + \frac{1}{\text{a}-\text{x}}\right) + \frac{\text{kq}^2}{2\text{a}}$

Hence, $\mathrm{k} = \frac{1}{4\pi\varepsilon_0}$

$|\Delta \mathrm{U}| = |\mathrm{U_f} - \mathrm{U_i}| = \frac{2\mathrm{kQqx}^2}{\mathrm{a}^3}$ (for $\mathrm{x} << \mathrm{a}$)

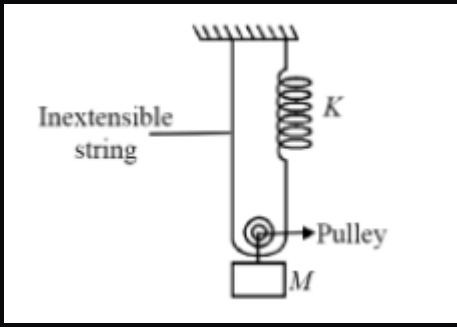
$\therefore \Delta \mathrm{U}\ \alpha\ \mathrm{x}^2$

Q.16 Answer:

$2\pi\sqrt{\frac{M}{4k}}$

Solution:

When mass M is



suspended from the given system as shown in figure, let l be the length through which mass M moves down before it comes to rest.

In this situation, both the spring and string will be stretched by length l. Since string is inextensible, so spring is stretched by length $2l$. The tension along the string and spring is the same.

In equilibrium, $Mg = 2(k2l)$ If mass M is pulled down through small distance x, then

$F = Mg - 2k(2l + 2x) = -4kx \ldots$ (i)

$F \propto x$ and $-$ve sign shows that it is directed towards mean position. Hence, the mass executes simple harmonic motion.

For SHM, $F = -k'x \ldots$ (ii)

Comparing (i) and (ii), we get $k' = 4k$

$\therefore$ Time period, $T = 2\pi\sqrt{\frac{M}{4k}}$

Q.17 Answer:

modulus of elasticity.

Solution:

The dimension of moment of force (torque) is,

$$\tau = \text{distance} \times \text{force}$$
$$= \left[\mathrm{L}^1\right]\left[\mathrm{M}^1\,\mathrm{L}^1\,\mathrm{T}^{-2}\right]$$
$$= \left[\mathrm{M}^1\,\mathrm{L}^2\,\mathrm{T}^{-2}\right]$$

The dimension of surface tension is,

$$\mathrm{S} = \frac{\text{force}}{\text{length}}$$
$$= \frac{\left[\mathrm{M}^1\,\mathrm{L}^1\,\mathrm{T}^{-2}\right]}{\left[\mathrm{L}^1\right]}$$
$$= \left[\mathrm{M}^1\,\mathrm{T}^{-2}\right]$$

The dimension of modulus of elasticity is,

$$\mathrm{E} = \frac{\text{Stress}}{\text{Strain}} = \frac{\left(\frac{\text{force}}{\text{area}}\right)}{\text{strain}}$$
$$= \frac{\left(\frac{\left[\mathrm{M}^1\,\mathrm{L}^1\,\mathrm{T}^{-2}\right]}{\left[\mathrm{L}^2\right]}\right)}{\left[\mathrm{M}^0\,\mathrm{L}^0\,\mathrm{T}^0\right]}$$
$$= \left[\mathrm{M}^1\,\mathrm{L}^{-1}\,\mathrm{T}^{-2}\right]$$

The dimension of the coefficient of viscosity is,

$$\eta = \frac{\text{force} \times \text{length}}{\text{area} \times \text{velocity}}$$
$$= \frac{\left[\mathrm{M}^1\,\mathrm{L}^1\,\mathrm{T}^{-2}\right]\left[\mathrm{L}^1\right]}{\left[\mathrm{L}^2\right]\left[\mathrm{L}^1\,\mathrm{T}^{-1}\right]}$$
$$= \left[\mathrm{M}^1\,\mathrm{L}^{-1}\,\mathrm{T}^{-1}\right].$$

Q.18 Answer:

$mg\left[\frac{b-a}{2}\right]$

Solution:

Initial height of $\mathrm{CG} = \frac{a}{2}$

Final height of $\mathrm{CG} = \frac{b}{2}$

Work done $= mg\left[\frac{b}{2} - \frac{a}{2}\right] = mg\left(\frac{b-a}{2}\right)$

Q.19 Answer:

$\frac{EL}{eR^2}$

Solution:

In $L - R$ **circuit, the growing current at time** t **is given by**

$$i = i_0\left[1 - \mathrm{e}^{-t/\tau}\right] \quad \text{where, } i_0 = \frac{E}{R} \text{ and } \tau = \frac{L}{R}$$

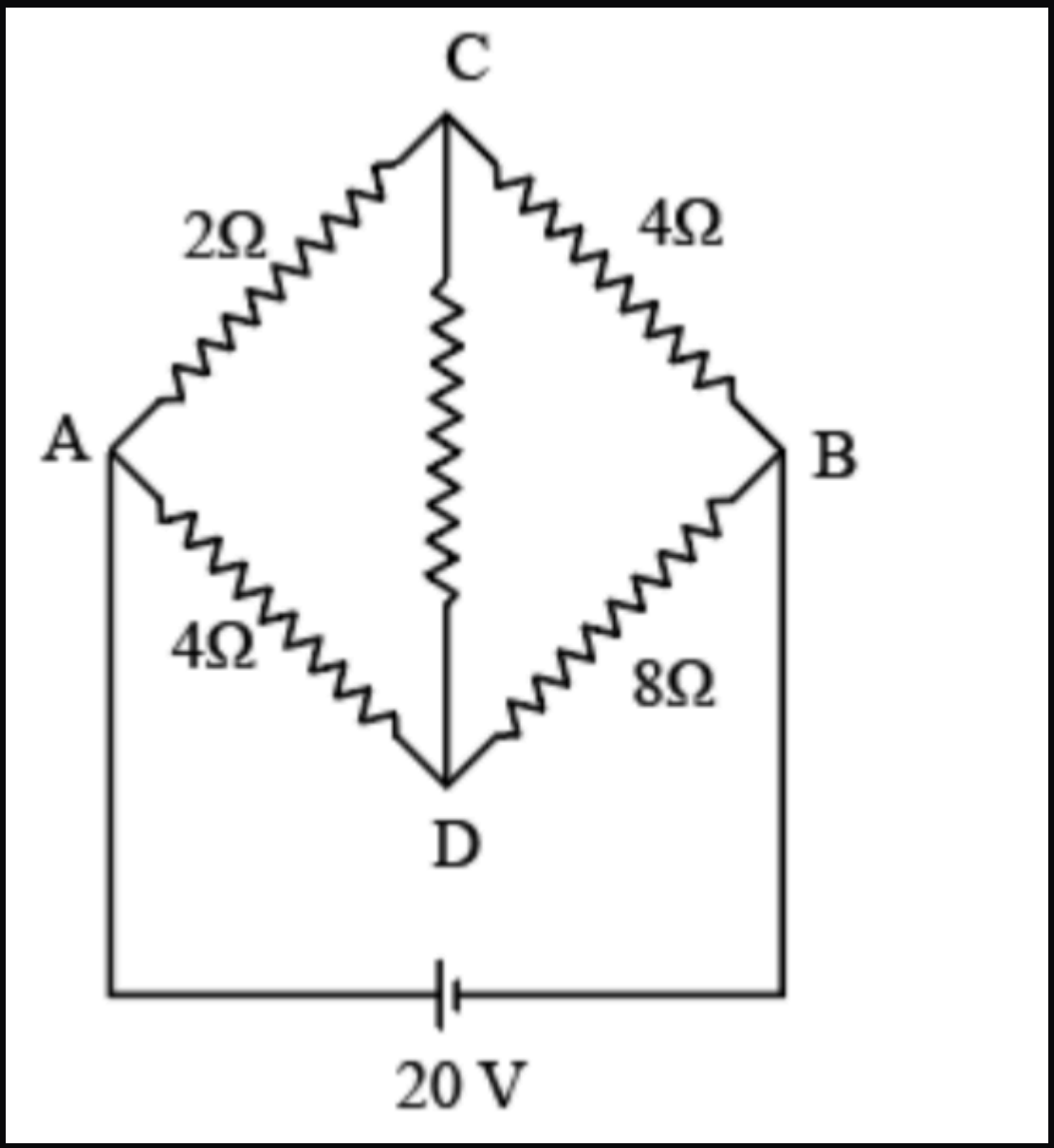
$\therefore$ Charge passed through the battery in one time constant is

$$q = \int_0^t i dt = \int_0^t i_0 \left(1 - e^{-1/\tau}\right) dt$$
$$q = i_0\tau - \left[\frac{i_0 e^{-t/\tau}}{-2/\tau}\right]_0^t = i_0\tau + i_0\tau\left[e^{-1} - 1\right]$$
$$= i_0\tau - i_0\tau + \frac{i_0\tau}{e}$$
$$q = \frac{i_0\tau}{e} = \frac{(E/R)(L/R)}{e} = \frac{El}{eR^2}$$

Q.20 Answer:

Current through the battery is $5\ \text{A}$

Solution:



By using the concept of Wheatstone bridge

$$V_{\mathrm{D}} - V_{\mathrm{C}} = 0$$

$$\therefore R_{\mathrm{eq}} = \frac{12 \times 6}{12 + 6} = 4\Omega$$

$$\therefore I_{\mathrm{battery}} = \frac{20}{4} = 5\ \mathrm{A}$$

Q.21 Answer:

—

Solution:

Speed of wave on a string $v = \sqrt{\frac{T}{m}}$

$\Rightarrow v \propto \sqrt{T} \Rightarrow \frac{v_1}{v_2} = \sqrt{\frac{T_1}{T_2}} \Rightarrow \frac{T_1}{T_2} = \left(\frac{v_1}{v_2}\right)^2$

$\Rightarrow \frac{T_2 - T_1}{T_1} = \frac{v_2^2 - v_1^2}{v_1^2}$

Given, $\mathrm{T} = 120\ \mathrm{m}, \mathrm{v}_1 = 150\ \mathrm{m/s}$

$v_2 = v_1 + \frac{20}{100} v_1 = \frac{120}{100} v_1 = \frac{6}{5} v_1 = \frac{6}{5} \times 100 = 180\ \mathrm{m/s}$

Substituting the values in eq. (i), we get,

$\frac{T_2 - T_1}{T_1} = \frac{(180)^2 - (150)^2}{(150)^2} = \frac{30 \times 330}{150 \times 150} = 0.44$

Percent increase in tension $= 44\%$

Q.22 Answer:

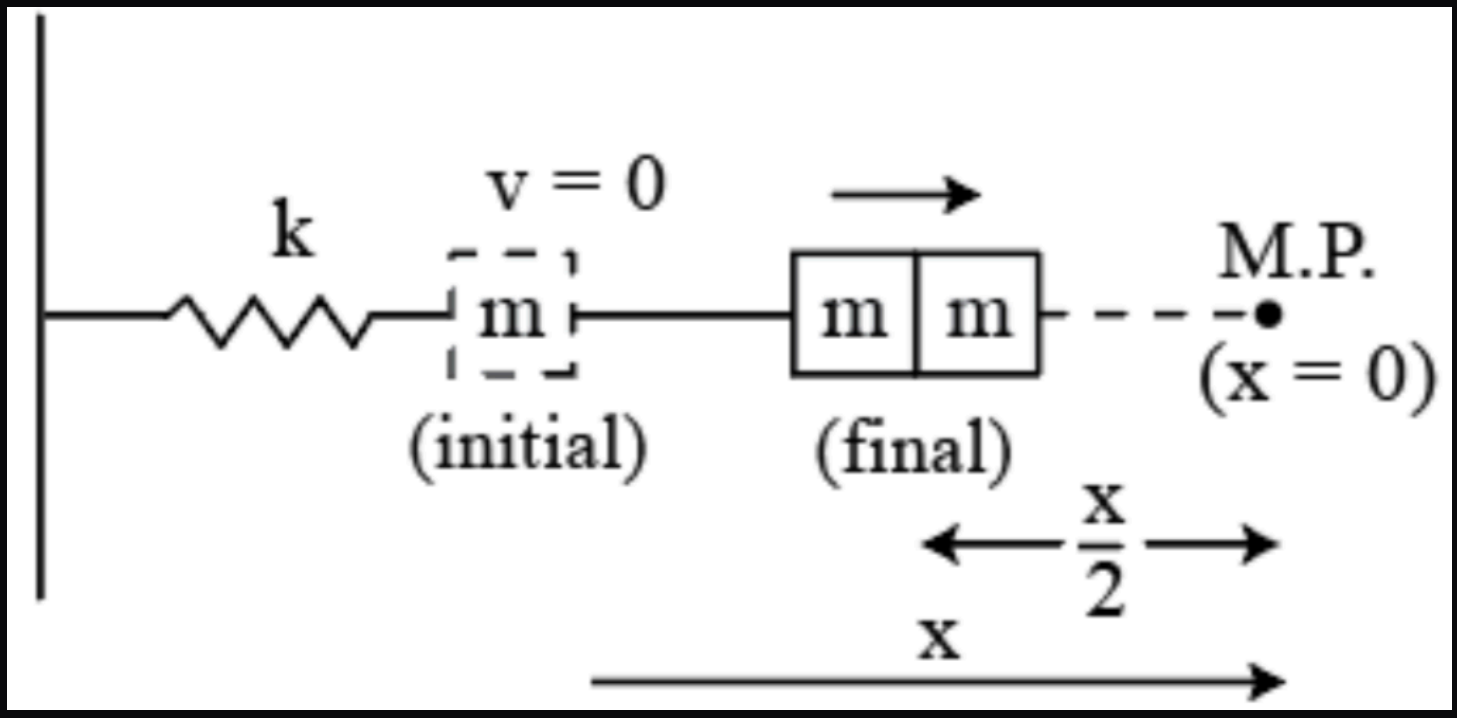
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Solution:

Momentum conservation gives $mu = (2m)(2)$

$$\therefore \quad u = 4ms^{-1}; \quad E_i = E_f$$

$$\Rightarrow \quad \frac{1}{2}kx^2 = \frac{1}{2}k\left(\frac{x}{2}\right)^2 + \frac{1}{2}m(4)^2 \Rightarrow \frac{3}{8}kx^2 = 2.4 \quad \therefore \quad \frac{1}{2}kx^2 = 3.2J$$

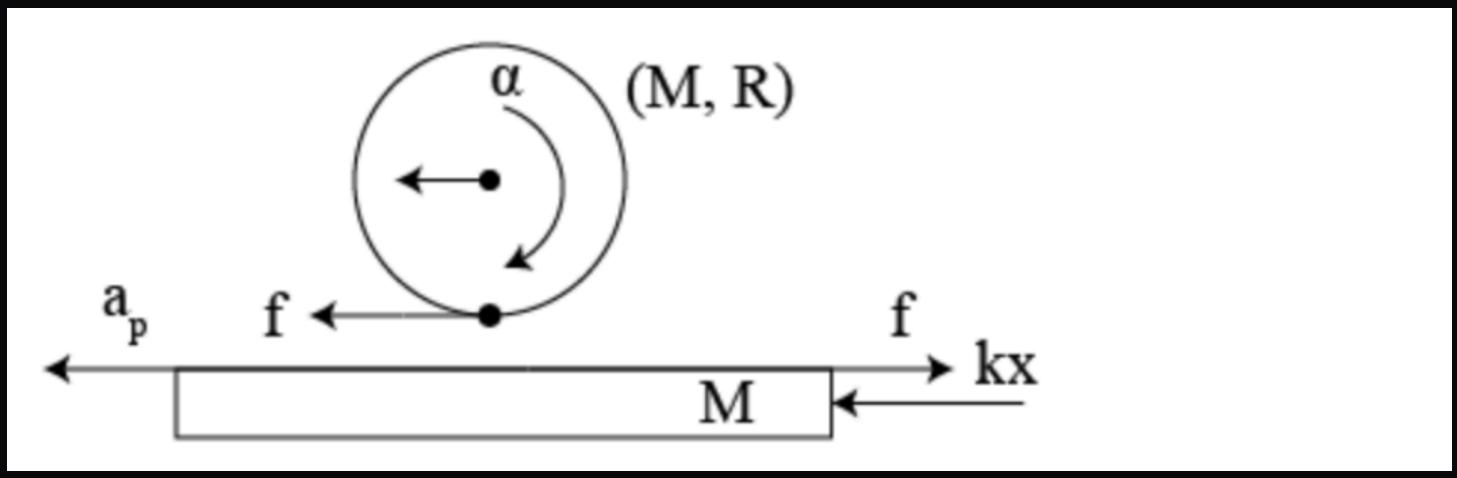


Q.23 Answer:

—

Solution:

Using no Slipping condition:



$a_c + \alpha R = a_p$

$\frac{f}{M} + \frac{2fR}{MR^2}R = \frac{kx-f}{M}$

$\Rightarrow 4f = kx$

$x_{\max} = \frac{4\mu Mg}{k}$

$\Rightarrow x_{\max} = \frac{4\times0.4\times2\times10}{200} = 0.16 \text{ m} = 16 \text{ cm}$

Q.24 Answer:

—

Solution:

Total 6 lines are emitted. Therefore,

$\frac{n(n-1)}{2} = 6$

or $n = 4$

So, transition is taking place between m^{th} **energy state and** $(\mathrm{m}+3)^{\mathrm{th}}$ **energy state.**

$E_m = -0.85\text{eV}$

Or

$-13.6\left(\frac{z^2}{m^2}\right) = -0.85$

Or

$\frac{z}{m} = 0.25\ldots \text{ (i)}$

Similarly, $E_m + 3 = -0.544\text{eV}$

or $-13.6\frac{z^2}{(m+3)^2} = -0.544$

$(\text{ii})\ \text{or}\ \frac{z}{(m+3)} = 0.2 \ldots (ii)$

(ii) or $\frac{z}{(m+3)} = 0.2 \ldots (ii)$

Solving Eqs. (i) and (ii) for z **and** m**, we get**

$m = 12 \text{ and } z = 3$

Q.25 Answer:

—

Solution:

Reading = MSR + CSR + Error

$0.322 = 0.300 + \text{CSR} + 5 \times \text{LC}$

$0.322 = 0.300 + \text{CSR} + 0.005$

$\text{CSR} = 0.017$

For (B)

Reading = MSR + CSR + Error

$0.322 = 0.200 + \text{CSR} + 0.092$

$\text{CSR} = 0.030$

Difference $= 0.030 - 0.017 = 0.013 \text{ cm}$

Division on circular scale $= \frac{0.013}{0.001} = 13$

Q.26 Answer:

$(\pm)$ 2, 3−butanediol

Solution:

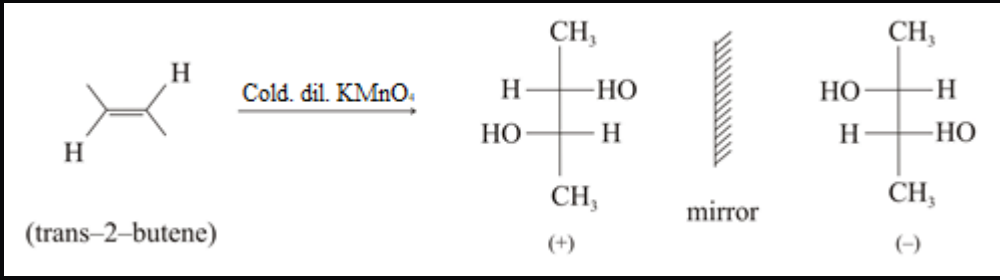
There are two types of addition reaction on alkenes.

1. Syn addtition: In this addition, groups or atoms are attacked from the same side on the reactant. When they attack, and there is cis alkene, then erythro product is formed in which both the attacking atoms and molecules are connected in same side while, in trans alkene threo product is formed in which both the attacking atoms and molecules are connected in opposite side. E.g. OsO_4, dil. $KMnO_4$

2. Anti addition: In this addition, groups or atoms are attacked from the opposite side on the reactant. When they attack, and there is trans alkene, then erythro product is formed in which both the attacking atoms and molecules are connected in same side while, in cis alkene threo product is formed in which both the attacking atoms and molecules are connected in opposite side. E.g. Br_2**, etc.**

If we talk about stereo chemistry then threo basically forms racemic mixture but erythro can form meso and racemic depends on structure that if it has plane of symmetry then it has meso and if it does not have plane of symmetry then it will be racemic structure.

So, following above theory we can say this given trans alkene will form racemic mixture as shown below:

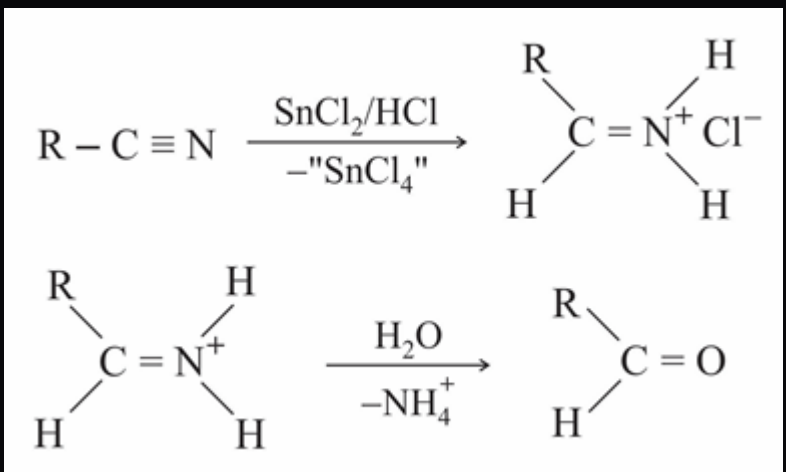


Q.27 Answer:

Aldehyde

Solution:

An alkyl cyanide is dissolved in ether or better in ethyl formate or ethyl acetate, and reduced with $SnCl_2$ **and** HCl **and then steam distilled. The whole process is called Stephen reaction. In this process, alkyl cyanide is reduced to aldehyde.**



There is no analogous method for the preparation of ketones.

Q.28 Answer:

BI_3

Solution:

The order of strength of Lewis acid character for boron halides is,

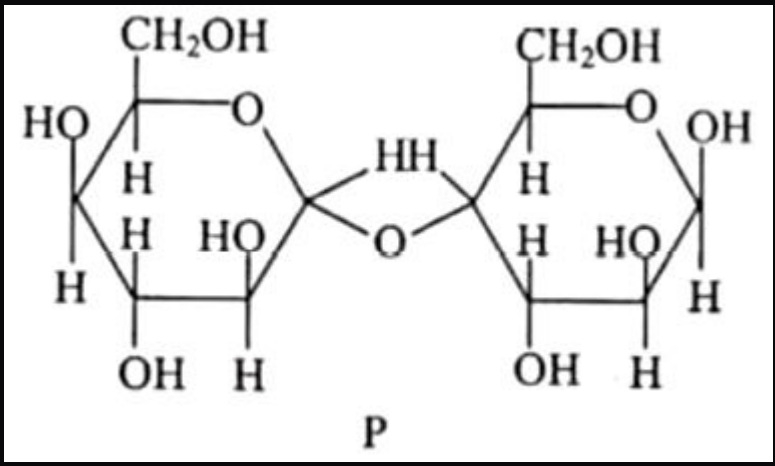
$BF_3 < BCl_3 < BBr_3 < BI_3$ (due to back bonding)

Q.29 Answer:

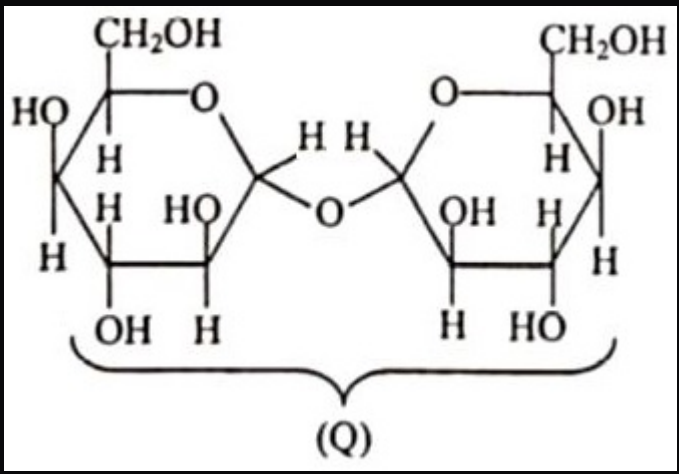
Q

Solution:

In (Q), it could be found that both the pyranose rings are attached through their reducing group, hence it will not be hydrolysed, and it can not reduce Tollen's reagent. So (Q) is not a reducing sugar, whereas in (P) one ring is hemiacetal which can be hydrolysed therefore it will be hydrolysed in solution and can reduce Tollen's reagent.



P

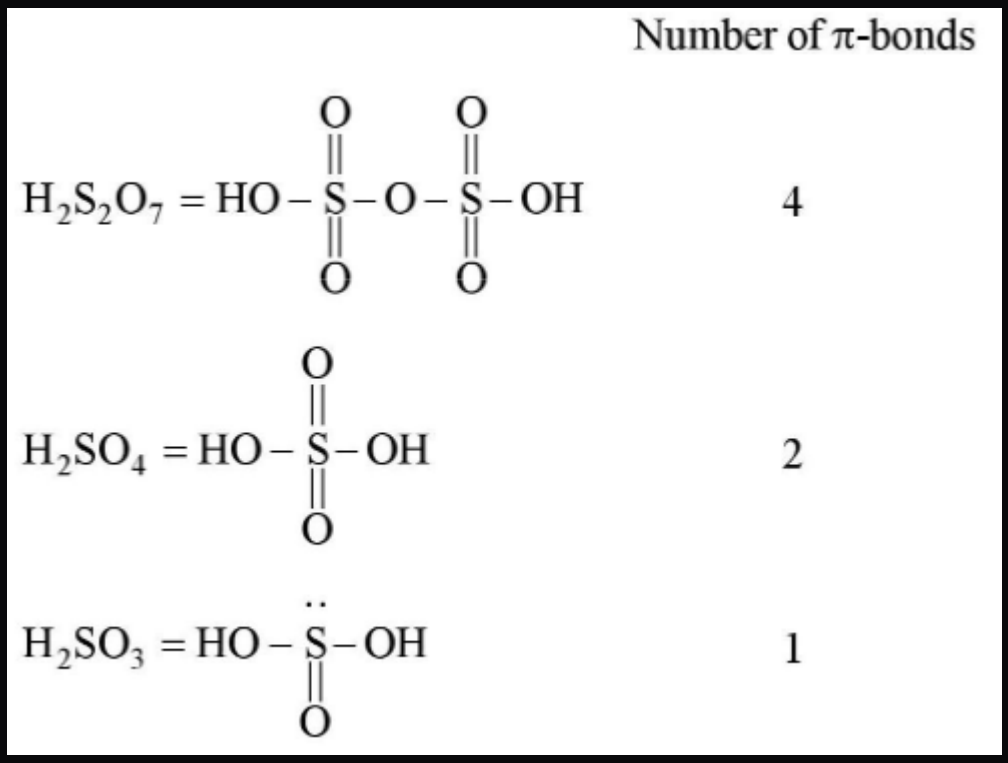


(Q)

Q.30 Answer:

$H_2\ S_2O_7 > H_2SO_4 > H_2SO_3$

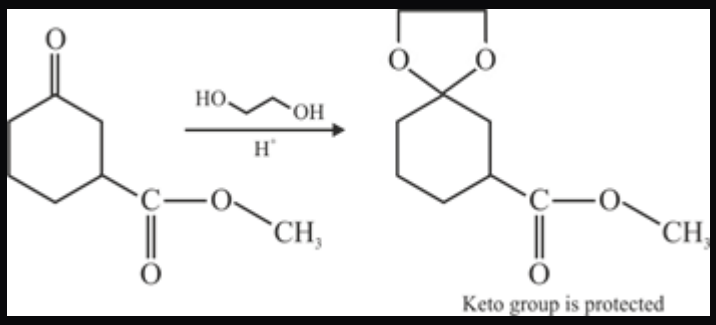
Solution:



Q.31 Answer:

$HOCH_2 - CH_2OH|H^+, 2CH_3MgBr, H_3O^+$

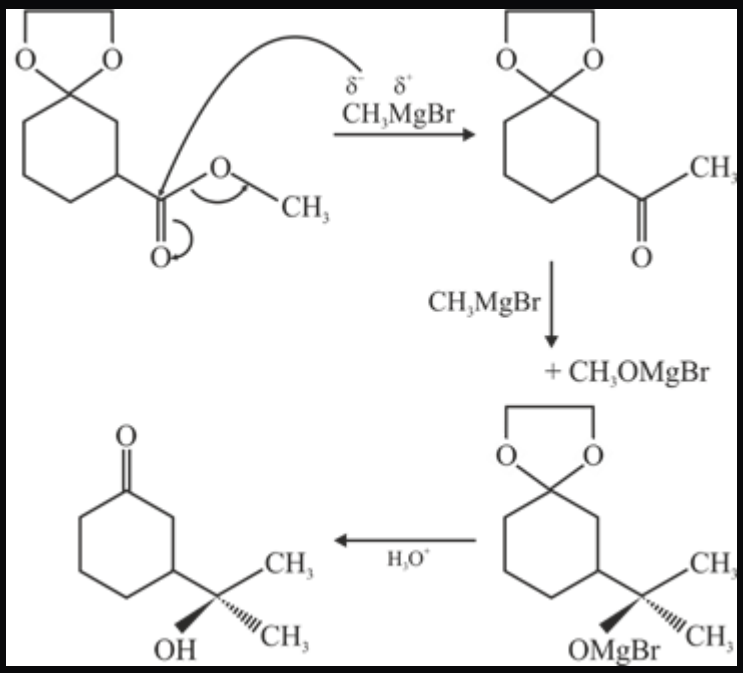
Solution:



Keto group is protected

Addition of Grignard reagent to ester group leads to formation of ketone and then alcohol after

hydrolysis.



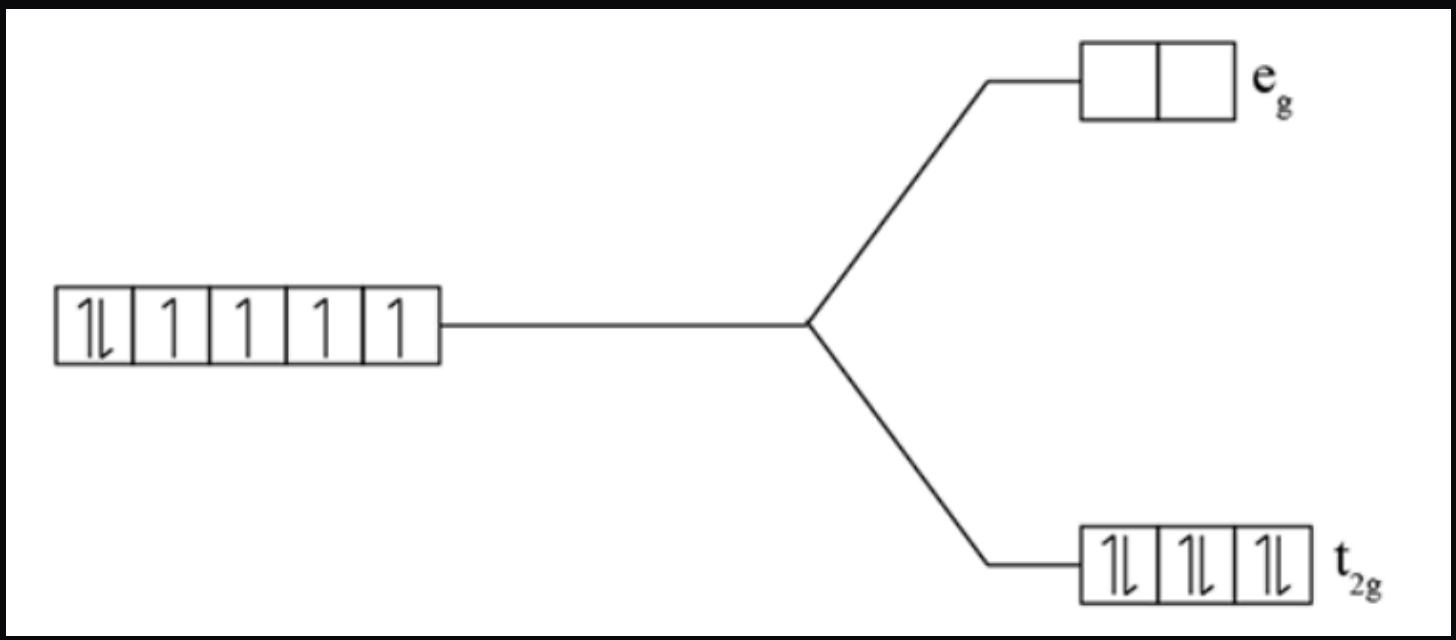
Q.32 Answer:

d-d transition

Solution:

When oxygen is bound to Fe^{2+} in heme, all the d-electrons of the Fe atom are forced into the three lower energy t_{2g} orbitals and it produces low spin oxyhaemoglobin.

In oxyhaemoglobin, $Fe^{2+} = [Ar]3d^6$



The sharp high energy transition between t_{2g} and e_g levels of the $d-$ electrons is responsible for the bright red colour of oxyhaemoglobin.

Q.33 Answer:

$Yb^{3+} < Pm^{3+} < Ce^{3+} < La^{3+}$

Solution:

Lanthanide contraction decreases atomic sizes from La^{3+} to Lu^{3+}, hence (a) will be correct order.

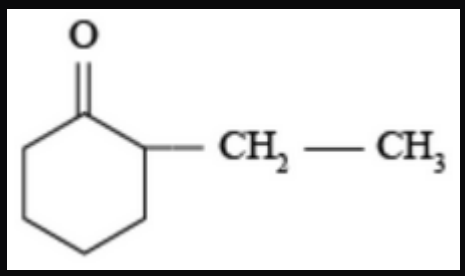
Q.34 Answer:

Assertion is true, but the reason is false.

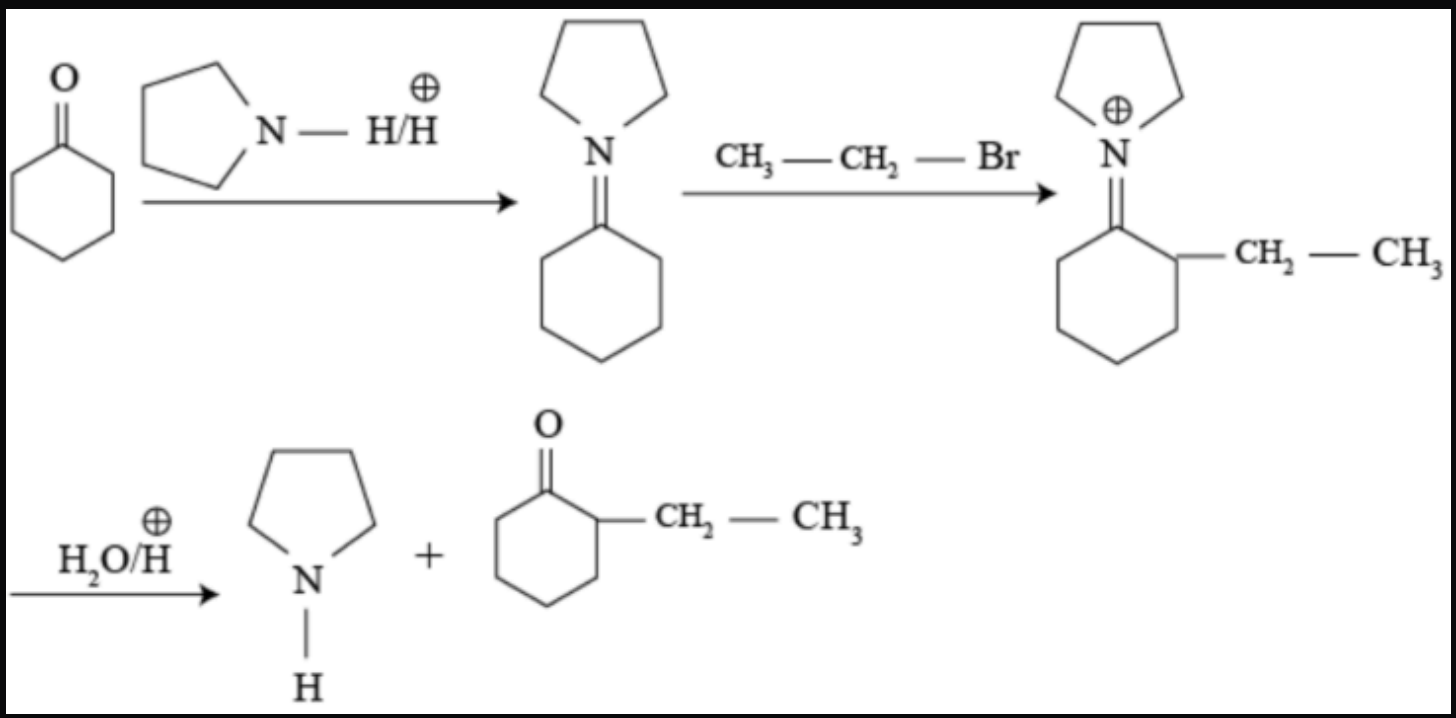
Solution:

Petroleum can be refined by the fractional distillation since it separates crude petroleum into useful fractions such as gasoline, kerosene oil, diesel oil, lubricating oil, etc.,

Q.35 Answer:



Solution:



Q.36 Answer:

$N(CH_3)_2H > N(CH_3)H_2 > N(CH_3)_3 > NH_3$

Solution:

In aqueous phase, the solvation of NH_4^+ ions with water should be considered. The energy of solvation contributes to their stability. Greater the size of ion, lesser is the solvation and lesser is the stability of ion. In secondary and tertiary amines, due to the steric hindrance, the alkyl groups decrease the number of water molecules that can approach closely and solvate the ions. Thus, the order of basicity of the aliphatic amines should be primary amines $>$ secondary amines $>$tertiary amines. To compare the basicity of the amines, the inductive effect, the solvation effect, the steric hindrance etc., should be taken into consideration. The order of basic strength in aqueous phase is as follows:

$$N(CH_3)_2H > N(CH_3)H_2 > N(CH_3)_3 > NH_3.$$

Q.37 Answer:

Rest mass of photon is zero

Solution:

(i) Fact

(ii) Unit of plank's constant is Joule second

(iii) Splitting of spectral lines in magnetic field is called Zeeman effect

(iv) Bohr's model is applicable only to single electron species

Q.38 Answer:

CH_3COCH_3

Solution:

Initially, the carbyl amine reaction takes place and forms isocyanide, which is a foul smelling gas. Y would be chloroform ($CHCl_3$). According to the given option, Z would be acetone as it can give the chloroform reaction.

$$Ph-NH_2 \xrightarrow[CHCl_3]{KOH} Ph-NC$$

$$CH_3COCH_3 \xrightarrow{Cl_2\,/\,Ca(OH)_2} (CH_3COO)_2Ca + CHCl_3$$

Q.39 Answer:

I, II and III only

Solution:

I $\log(a-x)$ (where $a-x$ is the remaining concentration), decreases with the passage of time.

II $kt = 2.303 \log\left(\frac{a}{a-x}\right)$

III For first order $t_{\frac{1}{2}}$ is independent of a

Q.40 Answer:

2

Solution:

For the reaction:

$$\mathrm{NH_4Cl(s) \rightleftharpoons NH_3(\ g) + HCl(g)}$$

$\Delta \mathrm{n_g}$ = Number of moles of gaseous product − Number of moles of reactant

$$\Delta \mathrm{n_g} = 2 - 0$$

$$\Delta \mathrm{n_g} = 2$$

Q.41 Answer:

$\mathrm{Ba^{2+} > Cs^{+} > Se^{2-} > As^{3-}}$

Solution:

In (1) and (2) use $(\mathrm{z/e})$ concept for isoelectronic species. In (3) size of neutral atom is greater than its cation.

Q.42 Answer:

4.08%

Solution:

Let the no. of moles of $\mathrm{M^{2+}}$ be x

$$\begin{aligned} \therefore \quad & 2x + (0.98 - x) \times 3 = 2 \\ & x = 0.94 \\ \therefore \quad & M^{3+} \text{ present } = 0.04 \\ \% \quad & M^{3+} = \frac{0.04}{0.98} \times 100 = 4.08\% \end{aligned}$$

Q.43 Answer:

40ml

Solution:

$[H_2CO_3]$ **in blood** $= 2M$

Volume of blood $= 10ml$

$[NaHCO_3] = 5M$

Let volume of $NaHCO_3$ **used** $= Vm$

$[H_2CO_3]$ **in mixture** $= \frac{2\times10}{(v+10)}$

$[NaHCO_3]$ **in mixture** $= \frac{(5\times v)}{(v+10)}$

$pH = pKa + \log\frac{(\text{ salt })}{(\text{ acid })}$

$7.4 = -\log\left(4\times10^{-7}\right) + \log\frac{\frac{(5\times v)}{(v+10)}}{\frac{(2\times10)}{(v+10)}}$

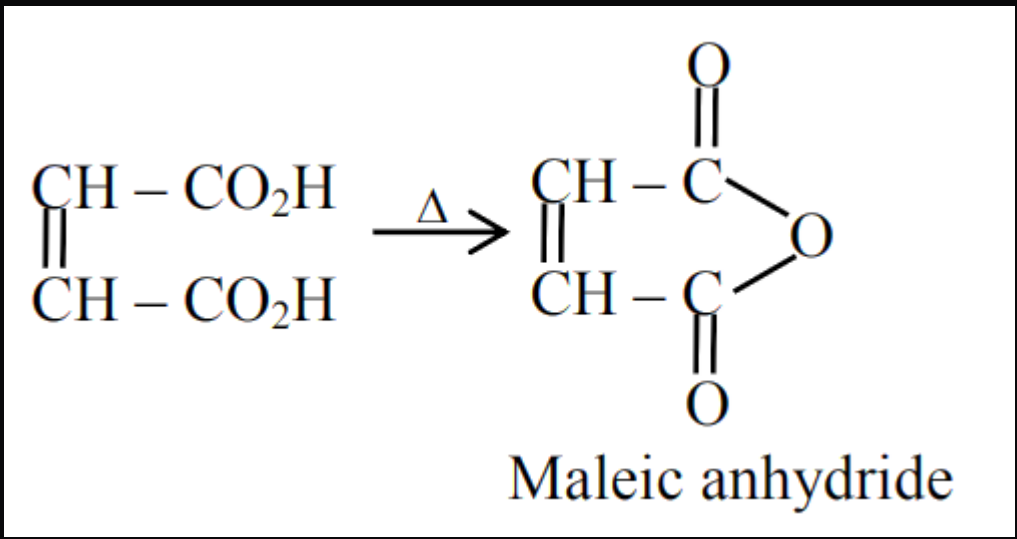
$v = 40ml$

Q.44 Answer:

Heating cis-but-2-enedioic acid

Solution:

Heating cis-but-2-enedioic acid give maleic anhydride as follows,



Q.45 Answer:

II $>$ I $>$ III

Solution:

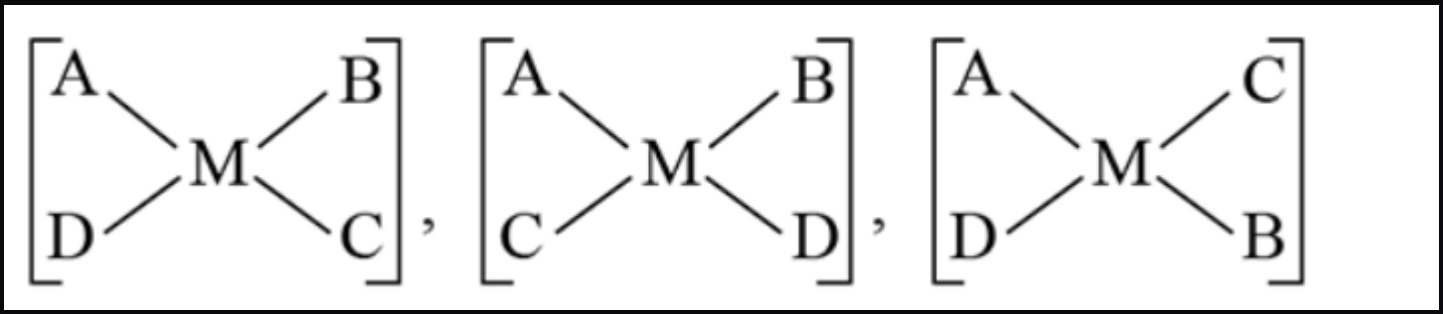
All three given carbocations are 2° **carbocations. There is** $+I$ **and** $+M$ **effect due to** OCH_3 **that stabilise carbocation in (II). There is - I effect due to** $COCH_3$ **that destabilizes carbocation in (I). Order of stability will be** $II > I > III$

Q.46 Answer:

—

Solution:

Complex is square planar



Q.47 Answer:

—

Solution:

Let order of reaction is x

$$\frac{-dA}{dt} = K[A]^x$$

On integrating and using limits A_0 to $\frac{A_0}{2}$ and 0 to t for time

$$\frac{\left(\frac{A_0}{2}\right)^{-x+1} - (A_0)^{-x+1}}{1-x} = -kt$$
$$A_0^{-x+1}\frac{\left(1-2^{-1+x}\right)}{1-x} = kt$$

Put $t = 100$ and $A_0 = 0.1$

$$(0.1)^{-x+1}\frac{\left(1-2^{-1+x}\right)}{1-x} = k(100) \ldots (1)$$

Put $t = 50$ ond $A_0 = 0.025$

$$(0.025)^{-x+1}\left(\frac{1-2^{-1+x}}{1-x}\right) = K(50)\ldots(2)$$

divide (1) by (2)

$$\left(\frac{0.1}{0.025}\right)^{-x+1} = 2$$
$$\therefore \quad x = 0.5$$

N=0.5, 4N=2

Q.48 Answer:

—

Solution:

If $4\text{fn} = 4, 1 = 3$

Radial nodes $(X) = n - 1 - 1 = 4 - 3 - 1 = 0$

Angular nodes $(Y) = 1 = 3$

$\mathrm{X + Y = 0 + 3 = 3}$

Q.49 Answer:

—

Solution:

Only symmetrical alkanes are prepared using wurtz reaction. So only $\mathrm{CH_3 - CH_3}$ and $\mathrm{CH_3 - \overset{CH_3}{\overset{|}{C}H} - \overset{CH_3}{\overset{|}{C}H} - CH_3}$ can be prepared using this reaction.

Q.50 Answer:

—

Solution:

Writing the formation reaction

$4C(s) + 2H_2(g) + \frac{1}{2}O_2(g) \rightarrow C_4H_4O(g)$

$\Delta H_f^\circ = -62 \cdot 0 = \Sigma_{\text{reacants}} \text{Enthalpy of atomisation} - \Sigma_{\text{products}} \text{Enthalpy of atomisation}$

$\Sigma_{\text{products}} \text{ Enthalpy of atomisation} = \Sigma_{\text{reactants}} \text{Enthalpy of atomisation} + 62 \cdot 0$

$= 4 \times 717 + 4 \times 218 + 249 + 62$

$= 2868 + 872 + 249 + 62 = 4051 \text{ kJ mol}^{-1}$

Q.51 Answer:

-7

Solution:

We have,

$\vec{a} = 2\hat{i} + \hat{k}$

$\vec{b} = \hat{i} + \hat{j} + \hat{k}$

$\vec{c} = 4\hat{i} - 3\hat{j} + 7\hat{k}$

Then,

$\vec{a} \cdot \vec{c} = \left(2\hat{i} + \hat{k}\right) \cdot \left(4\hat{i} - 3\hat{j} + 7\hat{k}\right) = 15$

$\vec{a} \cdot \vec{b} = \left(2\hat{i} + \hat{k}\right) \cdot \left(\hat{i} + \hat{j} + \hat{k}\right) = 3$

Now,

$\vec{r} \times \vec{b} = \vec{c} \times \vec{b}$

$\Rightarrow \vec{a} \times \left(\vec{r} \times \vec{b}\right) = \vec{a} \times \left(\vec{c} \times \vec{b}\right)$

$\Rightarrow \left(\vec{a} \cdot \vec{b}\right)\vec{r} - \left(\vec{a} \cdot \vec{r}\right)\vec{b} = \left(\vec{a} \cdot \vec{b}\right)\vec{c} - \left(\vec{a} \cdot \vec{c}\right)\vec{b}$

$\Rightarrow \left(\vec{a} \cdot \vec{b}\right)\vec{r} = \left(\vec{a} \cdot \vec{b}\right)\vec{c} - \left(\vec{a} \cdot \vec{c}\right)\vec{b}$

$\left[\because \left(\vec{a} \cdot \vec{r}\right) = 0\right]$

$\Rightarrow \left(\vec{a} \cdot \vec{b}\right)\vec{r} = \left(\vec{a} \cdot \vec{b}\right)\vec{c} - \left(\vec{a} \cdot \vec{c}\right)\vec{b}$

$\Rightarrow 3\vec{r} = 3\vec{c} - 15\vec{b}$

$\Rightarrow \vec{r} = \vec{c} - 5\vec{b}$

$\Rightarrow \vec{r} = \left(4\hat{i} - 3\hat{j} + 7\hat{k}\right) - 5\left(\hat{i} + \hat{j} + \hat{k}\right)$

$\Rightarrow \vec{r} = \left(-\hat{i} - 8\hat{j} + 2\hat{k}\right)$

Then,

$\vec{r} \cdot \vec{b} = \left(-\hat{i} - 8\hat{j} + 2\hat{k}\right) \cdot \left(\hat{i} + \hat{j} + \hat{k}\right)$

$= -1 - 8 + 2 = -7.$

Q.52 Answer:

-1624

Solution:

$\therefore \text{adj}(adjA) = |A|A$ **(by property)**

$A\ adj(adj\ A) = |A|A^2 \ldots$ **(i)**

$$|A| = \begin{vmatrix} 2 & 1 & -1 \\ 3 & 5 & 2 \\ 1 & 6 & 1 \end{vmatrix} = 2(5-12) - 1(3-2) - 1(18-5)$$

$$= -14 - 1 - 13 = -28$$

$$A^2 = \begin{bmatrix} 2 & 1 & -1 \\ 3 & 5 & 2 \\ 1 & 6 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 & -1 \\ 3 & 5 & 2 \\ 1 & 6 & 1 \end{bmatrix} = \begin{bmatrix} 6 & 1 & -1 \\ 23 & 40 & 9 \\ 21 & 37 & 12 \end{bmatrix}$$

$$\text{tr}(A(adj(adjA))) = \text{tr}\left(|A|A^2\right) = -28[6 + 40 + 12]$$

$$= -28 \times 58 = -1624$$

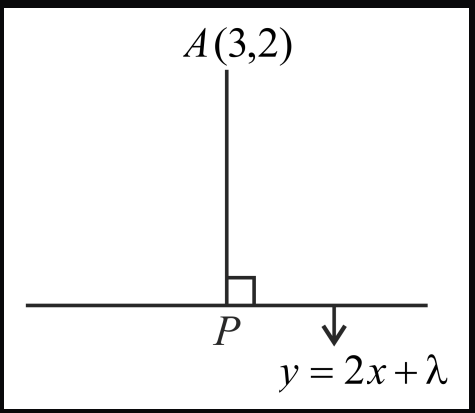
Q.53 Answer:

$x + 2y - 7 = 0$

Solution:

Equation of BC varies.

$\therefore$ Orthocentre will always lie on line perpendicular to $y = 2x + \lambda$ passing through $A(3, 2)$.



Hence, slope of AP is -2.

Equation of AP is

$y - 2 = -\frac{1}{2}(x - 3)$

$\Rightarrow x + 2y = 7$

$\Rightarrow$ Locus of orthocentre H is $x + 2y - 7 = 0$.

Q.54 Answer:

32

Solution:

For the G.P. $a, ar, ar^2, \ldots\ldots\ldots$

$P_n = a(ar)\left(ar^2\right)\ldots.\left(ar^{n-1}\right)$

$= a^n r^{\frac{n(n-1)}{2}}$

$\Rightarrow \sqrt[n]{P_n} = ar^{\left(\frac{n-1}{2}\right)}$

$\Rightarrow \sum_{n=1}^{\infty} \sqrt[n]{P_n} = a[1 + \sqrt{r} + r + r\sqrt{r} + \ldots\ldots\ldots]$

$= \frac{a}{1-\sqrt{r}}$

Given, $a = 16$ and $r = \frac{1}{4}$

$= \frac{16}{1-\sqrt{\frac{1}{4}}} = 32$

Q.55 Answer:

5

Solution:

$\sin^{-1}(\sin 12) + \cos^{-1}(\cos 12) = 0$

$(\mathrm{n}-2)\mathrm{x}^2 + 8\mathrm{x} + \mathrm{n} + 4 > 0; \forall \mathrm{x} \in \mathrm{R}$
$64 - 4(\mathrm{n}-2) \Rightarrow (\mathrm{n}+4) < 0 \text{ and } \mathrm{n}-2 > 0$
$\Rightarrow 16 - \left(\mathrm{n}^2 + 2\mathrm{n} - 8\right) < 0 \text{ and } \mathrm{n} > 2$
$\Rightarrow \mathrm{n}^2 + 2\mathrm{n} - 24 > 0 \text{ and } \mathrm{n} > 2$
$\Rightarrow (\mathrm{n}+6)(\mathrm{n}-4) > 0 \text{ and } \mathrm{n} > 2$
$\Rightarrow \mathrm{n} > 4 \text{ as } \mathrm{n} \in \mathrm{N} \text{ and } \mathrm{n} > 2$
$\Rightarrow \mathrm{n} \geq 5$

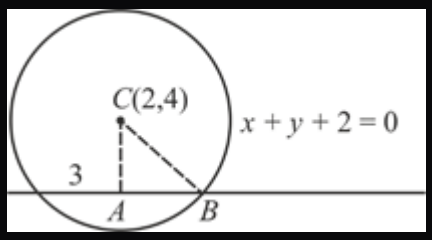
Hence, the least value of n is 5 .

Q.56 Answer:

$\sqrt{41}$ units

Solution:

Let, $CB = r$ be the radius of the circle.



Now, perpendicular distance

$AC = \frac{|2+4+2|}{\sqrt{1^2+1^2}} = \frac{8}{\sqrt{2}} = 4\sqrt{2}$

In right-angled ΔCAB,we have,

$CB^2 = r^2 = (AC)^2 + (AB)^2$

$\Rightarrow r^2 = (4\sqrt{2})^2 + (3)^2 = 32 + 9$

$\Rightarrow r^2 = 41$

$\therefore r = \sqrt{41}$ **units.**

Hence, the radius of the circle is $\sqrt{41}$ units.

Q.57 Answer:

720

Solution:

Case $(1) : d = 7$

${}^6C_3 \times {}^3C_3 = 20$

Case $(2) : d = 8$

${}^7C_3 \times {}^4C_3 = 140$

Case $(3) : d = 9$

${}^8C_3 \times {}^5C_3 = 560$

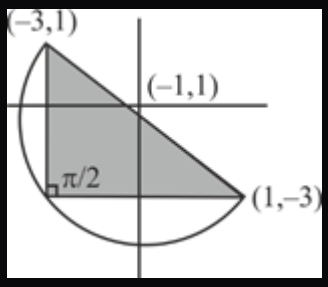
Total $= 20 + 140 + 560 = 720$

Q.58 Answer:

a part of circle with centre at the point $(-1, -1)$ and radius $2\sqrt{2}$

Solution:

$\arg\left(\frac{z-(-3+i)}{z-(1-3i)}\right)=\frac{\pi}{2}$



So, the required locus is shown in diagram with centre $(-1,-1)$ and radius $2\sqrt{2}$.

Q.59 Answer:

$\frac{1}{8}$

Solution:

$3^{2n}=9^n=(1+8)^n$

$\Rightarrow 3^{2n}={}^nC_0+{}^nC_1\cdot 8+{}^nC_2\cdot 8^2+\ldots..+{}^nC_n\cdot 8^n$

$\Rightarrow \frac{3^{2n}}{8}=\frac{{}^nC_0+{}^nC_1\cdot 8+{}^nC_2\cdot 8^2+.....+{}^nC_n\cdot 8^n}{8}$

$\Rightarrow \frac{3^{2n}}{8}=\frac{1}{8}+({}^nC_1+{}^nC_2\cdot 8+\ldots...+{}^nC_n\cdot 8^{n-1})$

$\Rightarrow \frac{3^{2n}}{8}=\frac{1}{8}+$ **integer quantity**

$\{x\}\in[0,\ 1)\ \&\ \{\mathrm{I}\}=0$

$\Rightarrow \quad \left\{\frac{3^{2n}}{8}\right\}=\frac{1}{8}+0$

Q.60 Answer:

$(-7,20)$

Solution:

Let $f(x)=2x^3-3x^2-12x+a$

Then,

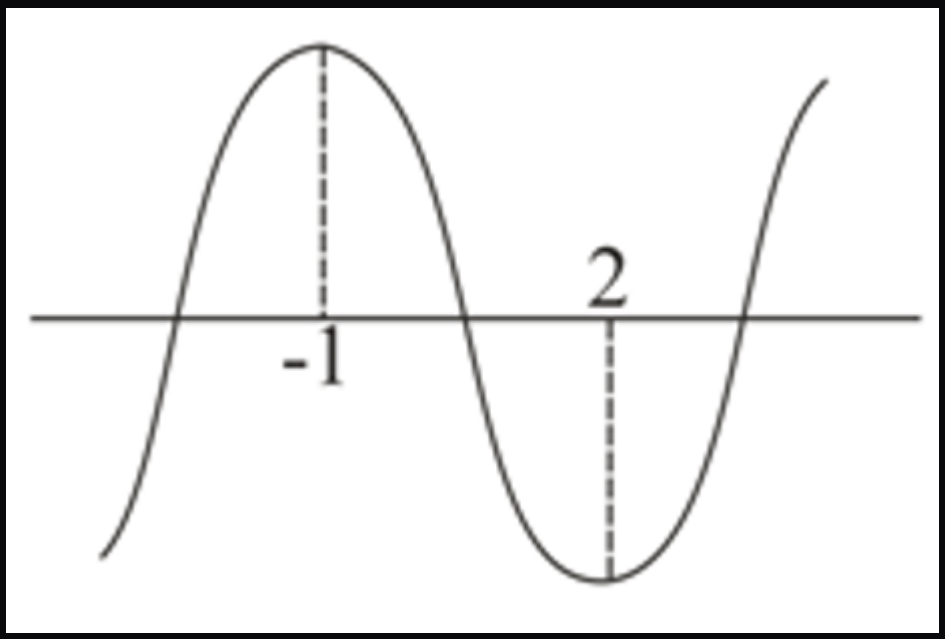
$$f'(x)=6\left(x^2-x-2\right)$$

$$\left\{\because \frac{d}{dx}\left(x^n\right)=nx^{n-1}\right\}$$

$$f'(x)=6(x+1)(x-2)$$

So, the roots of

$$f'(x) = 0 \text{ are } x = -1, 2.$$



Now, $f(x) = 0$ will have all real roots, if $f(-1) > 0$ and $f(2) < 0$.

Now, $f(-1) > 0$

$$\Rightarrow -2 - 3 + 12 + a > 0$$
$$\Rightarrow a > -7$$

and $f(2) < 0$

$$\Rightarrow 16 - 3(4) - 24 + a < 0$$
$$\Rightarrow a < 20$$

Hence, $-7 < a < 20$.

Q.61 Answer:

85

Solution:

In the nth set of observation, the total observations are 'n'

and the first observation of nth set is

$1+2+3+\ldots.+(n-1)+1=\frac{(n-1)n}{2}+1=a$

Sum of all the n observations in the nth set

$=\frac{n}{2}[2a+(n-1)d]$

$=\frac{n}{2}\left[2\left\{\frac{(n-1)n}{2}+1\right\}+(n-1)\right]$

$=\frac{n}{2}\left[n^2-n+2+n-1\right]=n\left(\frac{n^2+1}{2}\right)$

Mean $=\frac{n\frac{\left(n^2+1\right)}{2}}{n}=\frac{n^2+1}{2}$

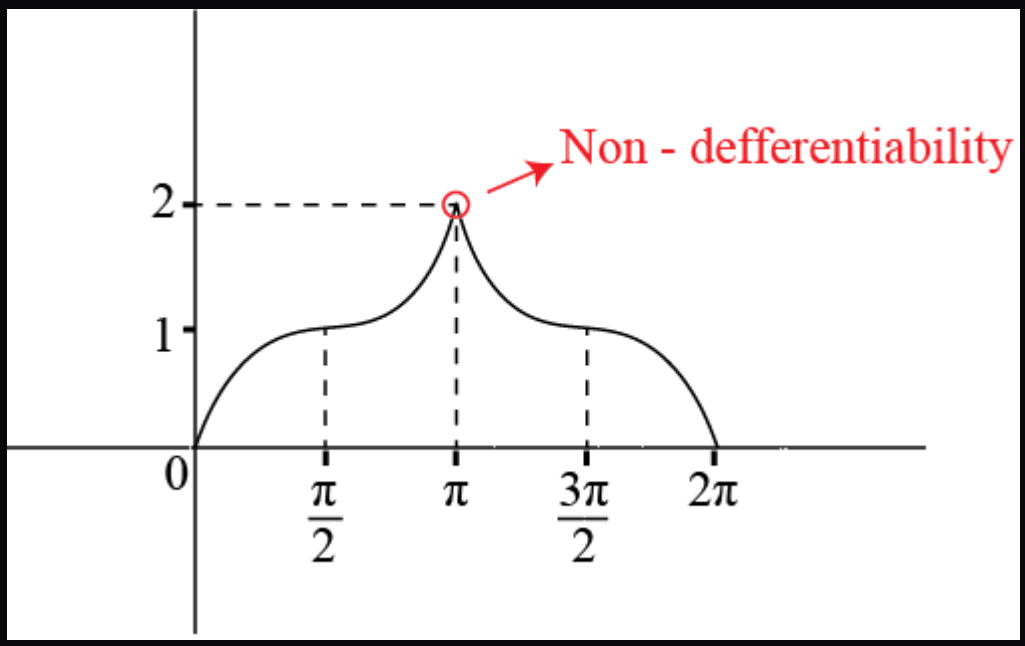
Hence, for $n=13$

Mean $=\frac{13^2+1}{2}=\frac{170}{2}=85$

Q.62 Answer:

1

Solution:



$f(x)=2-f(\pi-x)=2-\sin x$

$f(x)=f(2\pi-x)$

Continuous on $[0,2\pi)$

$$\Rightarrow n = 0$$
$$m = 1$$
$$n + m = 1$$

Q.63 Answer:

-2

Solution:

The midpoint of the segment is $\left(\frac{2-4}{2} \cdot \frac{3+1}{2}, \frac{2+4}{2}\right)$
$= (-1, 2, 3)$ **and the vector between this point and**
$(-3, 6,1)$ **is** $< -1 - (-3),\ 2 - 6.3 - 1 >$**and**
$< 2, -4, 2 >$
The line has parametric equations $x = -3 + 2t$ **,**$y = 6 - 4t$ **and**$z = 1 + 2t$
Solving each of these equations for t **and setting the expressions equal makes the equation.**
$\frac{x+3}{a} = \frac{y-6}{b} = \frac{z-1}{c} \Rightarrow \frac{x+3}{1} = \frac{y-6}{-2} = \frac{z-6}{1}$
So $1(-2) \cdot 1 - (1 - 2 + 1) = -2$

Q.64 Answer:

$\frac{1}{8}$

Solution:

$$\sin \frac{9\pi}{14} \sin \frac{11\pi}{14} \sin \frac{13\pi}{14} = \sin\left(\frac{\pi}{2} + \frac{\pi}{7}\right) \sin\left(\frac{\pi}{2} + \frac{2\pi}{7}\right) \sin\left(\frac{\pi}{2} + \frac{3\pi}{7}\right)$$

$$= \cos \frac{\pi}{7} \cos \frac{2\pi}{7} \cos \frac{3\pi}{7}$$

$$= \cos \frac{\pi}{7} \cos \frac{2\pi}{7} \cos\left(\pi - \frac{4\pi}{7}\right)$$

$$= -\cos \frac{\pi}{7} \cos \frac{2\pi}{7} \cos \frac{4\pi}{7}$$

$$= \frac{-\sin \frac{8\pi}{7}}{8 \sin \frac{\pi}{7}} = \frac{-\sin\left(\pi + \frac{\pi}{7}\right)}{8 \sin \frac{\pi}{7}}$$

$$= \frac{1}{8}$$

Q.65 Answer:

$\frac{1}{3}$

Solution:

$\because |A| = 3$

$$B = \begin{bmatrix} a_{11} & \frac{1}{2}a_{12} & \frac{1}{4}a_{13} \\ 2a_{21} & a_{22} & \frac{1}{2}a_{23} \\ 4a_{31} & 2a_{32} & a_{33} \end{bmatrix}$$

$$|B| = 8\begin{vmatrix} a_{11} & \frac{1}{2}a_{12} & \frac{1}{4}a_{13} \\ a_{21} & \frac{1}{2}a_{22} & \frac{1}{4}a_{23} \\ a_{31} & \frac{1}{2}a_{32} & \frac{1}{4}a_{33} \end{vmatrix} = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = |A| = 3$$

$$\left|\left(B^T\right)^{-1}\right| = \frac{1}{|B^T|} = \frac{1}{|B|} = \frac{1}{3}$$

Q.66 Answer:

$\frac{27}{31}$

Solution:

$A =$ **Shubham attends annual meet,** $P(A) = \frac{3}{4}$

$B =$ **Shikha attends annual meet**

$$P\left(\frac{\mathrm{B}}{\mathrm{A}}\right) = \frac{9}{10}$$

$$P\left(\frac{A}{B}\right) = \frac{\frac{3}{4}\times\frac{9}{10}}{P(A)\cdot P\left(\frac{\mathrm{B}}{\mathrm{A}}\right)+P\left(\bar{A}\right)P\left(\frac{\mathrm{B}}{\mathrm{A}}\right)} = \frac{27}{\frac{3}{4}\times\frac{9}{10}+\frac{1}{4}\times\frac{4}{10}} = \frac{27}{31}$$

Q.67 Answer:

None of these

Solution:

$$f\left(f\left(x\right)\right) = f\left(\frac{x}{(1+x^n)^{\frac{1}{n}}}\right)$$

$$= \frac{\frac{x}{(1+x^n)^{\frac{1}{n}}}}{\left(1+\frac{x^n}{1+x^n}\right)^{\frac{1}{n}}}$$

$$= \frac{x}{(1+2x^n)^{\frac{1}{n}}}$$

$$f\left(f\left(f\left(x\right)\right)\right) = f\left(\frac{x}{(1+2^n)^{\frac{1}{n}}}\right)$$

$$= \frac{\frac{x}{(1+2x^n)^{\frac{1}{n}}}}{\left(1+\frac{x^n}{1+2x^n}\right)^{\frac{1}{n}}}$$

$$= \frac{x}{(1+3x^n)^{\frac{1}{n}}}$$

and so on.

$$\Rightarrow \left(\text{fofo } \ldots \text{ of}\right)\left(x\right) = \frac{x}{(1+nx^n)^{\frac{1}{n}}} = g\left(x\right)$$

$$\int g\left(x\right)x^{n-2}dx = \int \frac{x^{n-1}}{(1+nx^n)^{\frac{1}{n}}}dx$$

$$\Rightarrow I\left(10, x\right) = \int \frac{x^9}{(1+10x^{10})^{\frac{1}{10}}}dx$$

Let $1 + 10x^{10} = t$

$\Rightarrow 100x^9dx = dt$

$\mathrm{I}\left(10, x\right) = \frac{1}{100}\int \frac{1}{\left(t^{\frac{1}{10}}\right)}dt$

$= \frac{1}{100}\left(\frac{t^{1-\frac{1}{10}}}{1-\frac{1}{10}}\right)$

$= \frac{1}{90}t^{\frac{9}{10}}$

$= \frac{1}{90}\left(1+10x^{10}\right)^{\frac{9}{10}}$

$\Rightarrow \mathrm{I}\left(10, 1\right) = \frac{(11)^{\frac{9}{10}}}{90}$

$\Rightarrow p = 11, m = 9, k = 10, q = 90$

$\Rightarrow 10k - m - q + p = 100 - 9 - 90 + 11 = 12$

Q.68 Answer:

2^{n-1}

Solution:

The value of $f(x)$ for any value of $x$will be either 1995 or 1996.

We know that:

Even+Even=Even

Odd+Odd(odd number of times) =Even

Even+Odd=Odd

Here, we have to find number of functions for which sum:$f(1) + f(2) + f(3)+\ldots\ldots\ldots\ldots f(n)$ is an odd integer

As we know that,

Now when we put values:$1,\ 2,\ 3,\ \ldots\ldots\ldots n-1$ in function f then we will get only two outcomes either 1995 or 1996.

Here, $f(1) + f(2) + f(3)+\ldots\ldots f(n-1)\ = odd\ or\ even$Now, to get required sum odd , $f(n)$ will be fixed whatever above is even or odd.

Since each of $\ f(1),\ f(2),\ \ldots f(n-1)$ can take two possibilities while $f(n)$ can take only one value.

Therefore total number of ways= $2.2.2.\ldots.(n-1)times\ =\ 2^{n-1}$.

Hence, required number of functions are 2^{n-1}.

Q.69 Answer:

$\frac{1}{(2x+y)^2} = x^2 + 1 + Ce^{x^2}$

Solution:

Let, $2x + y = t \Rightarrow \frac{dy}{dx} + 2 = \frac{dt}{dx}$

$\frac{dt}{dx} + xt = x^3t^3 \Rightarrow \frac{1}{t^3}\frac{dt}{dx} + \frac{1}{t^2}x = x^3$

Let, $\frac{1}{t^2} = u \Rightarrow \frac{-2}{t^3}\frac{dt}{dx} = \frac{du}{dx}$

$\frac{du}{dx} + (-2x)u = -2x^3$

$\mathrm{I.F.} = e^{-\int 2xdx} = e^{-x^2} \Rightarrow u.\, e^{-x^2} = \int e^{-x^2}\left(-2x^3\right)dx$

$\frac{e^{-x^2}}{(2x+y)^2} = -2\int e^{-x^2}.x^3dx$

$\frac{e^{-x^2}}{(2x+y)^2} = \int e^{-x^2}.x^2(-2x)dx$

Let, $-x^2 = v$

$-2xdx = dv \Rightarrow \frac{e^{-x^2}}{(2x+y)^2} = -\int e^v vdv$

$\frac{e^{-x^2}}{(2x+y)^2} + v\cdot e^v - e^v = C \Rightarrow \frac{e^{-x^2}}{(2x+y)^2} - x^2e^{-x^2} - e^{-x^2} = C$

$\frac{1}{(2x+y)^2} = \left(x^2+1\right) + Ce^{x^2}$

Q.70 Answer:

4. 00

Solution:

The sum of 50 observations $\sum_{i=1}^{50} x_i = 245$

Also, $\sum_{i=1}^{50} x_i^2 = 1401$

The variance of remaining observations is

$\sigma^2 = \frac{\sum x_i^2}{N} - \left(\frac{\sum x_i}{N}\right)^2$

$= \left(\frac{1401-\left(2^2+3^2+5^2\right)}{47}\right) - \left(\frac{245-(2+3+5)}{47}\right)^2$

$= \left(\frac{1401-38}{47}\right) - \left(\frac{235}{47}\right)^2$

$= \frac{1363}{47} - (5)^2$

$= 29 - 25 = 4$

Q.71 Answer:

—

Solution:

Let, $I = \int \frac{x^{pq-p-1}}{(x^p+1)^q}dx$

On taking x^p **common from denominator and simplifying**

$I = \int \frac{x^{-p-1}}{(1+x^{-p})^q}dx$

Let, $1 + x^{-p} = t$

$= \frac{-1}{p}\int \frac{dt}{t^q}$

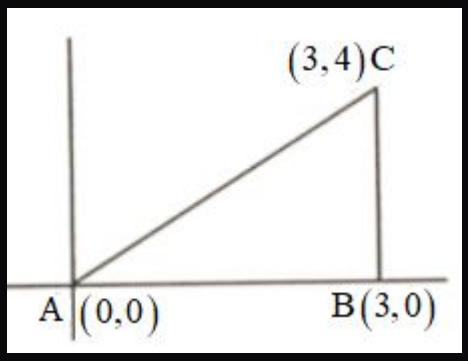
$= \frac{t^{-q+1}}{p(1-q)} + c$

$\therefore \ \frac{2}{\lambda} = 1 \ \Rightarrow \ \lambda = 2$

Q.72 Answer:

—

Solution:



$AC = 5 = 2ae$

$\Rightarrow 2ae = 5 \ldots (\text{i})$

Also $AB + BC = 2a$

$\Rightarrow 2a = 7 \Rightarrow a = \frac{7}{2}$

$\therefore e = \frac{5}{7}$

$\Rightarrow 1 - \frac{b^2}{a^2} = \frac{25}{49} \Rightarrow b = \sqrt{6}$

Length of the latus rectum, $\frac{2b^2}{a} = \frac{2 \times 6}{\frac{7}{2}}$ **units**

$\Rightarrow p = \frac{7}{2} = 3 \ .5$

Q.73 Answer:

—

Solution:

Given, $f(3x) = 2f(x)$

Also, $\int_0^1 f(x)dx = 1$

$\Rightarrow \frac{1}{2}\int_0^1 f(3x)dx = 1$

Substituting, $3x = t, \ dx = \frac{dt}{3}$**, we get,**

$\frac{1}{6}\int_0^3 f(t)dt = 1$

$\Rightarrow \int_0^3 f(t)dt = 6$

$\Rightarrow \int_0^1 f(t)dt + \int_1^3 f(t)dt = 6$

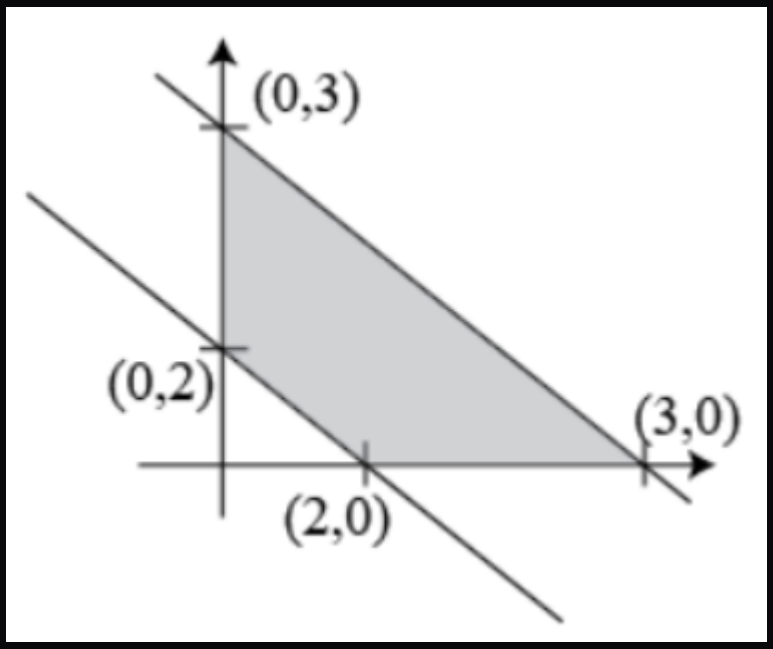
$\Rightarrow \int_1^3 f(t)dt = 6 - 1 = 5$

Q.74 Answer:

—

Solution:

As, $[x+y]=2 \Rightarrow x+y \in [2,3)$



The required area $= \left(\frac{1}{2} \cdot 3 \cdot 3\right) - \left(\frac{1}{2} \cdot 2 \cdot 2\right)$

$= \frac{9-4}{2} = \frac{5}{2}$ sq.units

Q.75 Answer:

—

Solution:

Unit vector $(\hat{n})$ perpendicular to $\overrightarrow{OA}$ and $\overrightarrow{CB}$

$\hat{n} = \frac{\overrightarrow{CB} \times \overrightarrow{OA}}{\left|\overrightarrow{CB} \times \overrightarrow{OA}\right|} = \frac{(3\alpha-7)\hat{i}-4\hat{j}+(5-\alpha)\hat{k}}{\sqrt{(3\alpha-7)^2+16+(5-\alpha)^2}}$

As $\overrightarrow{CB} \times \overrightarrow{OA} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & \alpha-3 & -1 \\ 1 & 2 & 3 \end{vmatrix} = \left|\overrightarrow{BA} \,.\, \hat{n}\right| = \left|\frac{(7-3\alpha)-4(2-\alpha)+2(5-\alpha)}{\sqrt{(3\alpha-7)^2+16+(5+\alpha)^2}}\right|$

Shortest distance $= \left|\overrightarrow{BA} \,.\, \overrightarrow{n}\right|$